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BVI 3215 – SYSTEM INTEGRATION DESIGNING

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FTKEE SMART MINI BOTTLING SYSTEM

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1.0 EXECUTIVE SUMMARY

Smart Mini Bottling System is a comprehensive industrial automation project focused on optimizing a small-scale liquid bottling process. The system is designed and implemented using automation technologies such as Programmable Logic Controllers (PLC), SCADA/HMI integration, and various sensors and actuators to ensure precision and efficiency. The project involves key components including PLC programming, instrumentation, electrical wiring, process design, and system commissioning. It is structured across three primary levels: Field Level (sensors and actuators), Control Level (PLC and logic control), and Supervisory Level (SCADA/HMI interface). Each team member plays a vital role in developing a fully functional smart bottling system, simulating a real-world industrial automation environment. The main objective is to design, develop, and commission a smart, automated bottling system that demonstrates the principles and practices of industrial process control and automation.

1.1 Objective and Key Roles

The main objective of the Smart Mini Bottling System project is to design and implement a small-scale, automated bottling process that mimics real-world industrial applications. This includes integrating PLC programming, SCADA/HMI monitoring, instrumentation, electrical wiring, physical process setup, and final system commissioning. Each team is assigned specific responsibilities to ensure the success of the project:

i. PLC Programming Team

- Develop and debug PLC code to control the entire bottling process.
- Collaborate with other teams to ensure proper signal and process integration.
- Test and validate the PLC logic for reliability and safety.

ii. SCADA/HMI Team

- Design intuitive and user-friendly interfaces for real-time process monitoring and control.
- Establish reliable communication between SCADA systems and PLCs.
- Enable visualization of critical data such as sensor readings, actuator status, and alarms.

iii. Instrumentation Team

- Select and install appropriate sensors (e.g., level, flow, pressure) for accurate monitoring.
- Calibrate all instrumentation to ensure precision and consistency.
- Validate the accuracy of sensor data used in automation control.

iv. Electrical Wiring Team

- Design and implement complete electrical wiring and panel layout.
- Ensure safe and proper distribution of power to all components.
- Conduct continuity checks and troubleshoot wiring issues.

v. Process Team

- Physically set up the mini plant structure, including tanks, pipes, and bottling mechanisms.
- Ensure proper mechanical installation and alignment of all components.
- Work closely with instrumentation and electrical teams to verify overall setup integrity.

vii. Commissioning Team

- Perform full system integration tests to ensure seamless operation.
- Detect and resolve issues during system start-up and tuning phases.
- Prepare final documentation and provide basic operational training to end-users.

1.2 Executive Summary: SCADA & HMI Team

The SCADA/HMI Team plays a critical role in the development and commissioning of the Smart Mini Bottling System, an industrial automation project centered around a simple chemical process utilizing the Mini Plant at FTKEE. The team is primarily responsible for the design, implementation, and integration of Supervisory Control and Data Acquisition (SCADA) and Human-Machine Interface (HMI) systems.

The key objective is to develop a user-friendly, responsive, and visually intuitive interface that enables real-time monitoring and control of the bottling process. This involves seamless communication with Programmable Logic Controllers (PLCs) to ensure efficient data flow and reliable system operation. The SCADA/HMI team focuses on:

- Designing clear and functional HMI screens.
- Implementing real-time data visualization and trend analysis.
- Facilitating smooth and robust communication between HMI and PLC systems.

1.2.1 SCADA & HMI Team – Key Responsibilities

The SCADA & HMI Team is responsible for the development and deployment of user-centric interfaces and robust communication systems that enable real-time monitoring and control of the Smart Mini Bottling System. Their key responsibilities are categorized as follows:

i. Interface Design and Implementation

- Design aesthetically pleasing, intuitive, and functional HMI screens tailored for operators and engineers.
- Develop SCADA systems that provide a comprehensive and centralized view of the entire industrial process.
- Ensure all interfaces are responsive and promote efficient user interaction with the system.

ii. Communication and Integration

- Establish and configure reliable communication protocols (e.g., Modbus, Ethernet/IP) between SCADA/HMI platforms and PLCs.
- Validate accurate and real-time data exchange to maintain seamless process control and monitoring.
- Collaborate actively with the PLC Programming Team to ensure full system integration and synchronization.

iii. Testing and Troubleshooting

- Conduct extensive testing of SCADA/HMI functionalities under various operational scenarios.
- Diagnose and resolve issues related to interface responsiveness, data lag, or system communication failures.
- Coordinate with all relevant teams to address integration or performance issues efficiently.

iv. User Training and Documentation

- Develop clear and concise user manuals and operation guides for system users and maintenance staff.
- Deliver hands-on training sessions to ensure all end-users are confident and capable in using the SCADA/HMI systems.
- Provide detailed documentation covering troubleshooting steps, routine maintenance, and system updates.

1.3 Recommendations for the Project

To ensure the successful integration and implementation of the Smart Mini Bottling System for the Mini Plant at FTKEE, several key recommendations are proposed. These focus on improving collaboration, system reliability, and overall project efficiency:

i. Collaborative Planning

- Foster continuous collaboration among all project teams throughout the entire development lifecycle.
- Schedule regular meetings and maintain open communication channels to quickly address any integration or technical challenges.

ii. Documentation Standardization

- Adopt standardized documentation formats and protocols across all teams.
- Consistent and organized documentation will enhance knowledge sharing, streamline troubleshooting, and support long-term system maintenance.

iii. User-Centric Design

- Apply user-centric design principles, particularly in the development of the SCADA/HMI interfaces.
- Engage end-users in the design and testing phases to gather feedback and improve interface usability and operator experience.

iv. Simulation and Testing

- Conduct thorough simulation and testing of all system components prior to commissioning.
- Simulations will help identify hidden issues early, reduce risks during deployment, and ensure smoother integration of all subsystems.

v. Timeline Flexibility

- Incorporate flexibility into the project schedule to accommodate unforeseen issues or design changes.
- A buffer in the timeline allows teams to make quality improvements without compromising deadlines.

vi. Regulatory Compliance

- Ensure the entire system complies with relevant industrial automation standards and safety regulations.
- Special attention should be given to electrical safety, software reliability, and process control integrity.

2.0 INTRODUCTION

2.1 Project Overview

The BVI3215 curriculum introduces the Mini Plant Project, a hands-on initiative designed to immerse students in the field of Industrial Automation System Integration using the FTKEE Mini Plant. Guided by experienced industry professionals and knowledgeable lecturers, students actively participate in the design, development, and implementation of an automation system for a basic chemical process.

This project serves as a practical platform for students to apply theoretical concepts in real-world-like settings. It emphasizes key aspects of industrial automation, including:

- Project preparation and planning
- Electrical wiring and power distribution
- Instrumentation and sensor calibration
- PLC programming and logic control
- SCADA/HMI interface development and integration

Beyond technical skills, the project encourages strong collaboration among team members, simulating the multidisciplinary teamwork found in actual industrial environments. The clearly defined roles and responsibilities offer students an in-depth understanding of each component within a fully integrated automation system.

Ultimately, the Mini Plant Project bridges the gap between academic knowledge and industrial practice. It equips students with the essential competencies needed for system commissioning and prepares them for the challenges and expectations of a career in industrial automation.

2.2 Objectives

The objectives of the Automated Bottling Station are designed to enhance operational efficiency, incorporate modern automation technologies, and facilitate effective control and monitoring of the system. These goals align with the overarching aim of bridging theoretical knowledge with real-world industrial applications.

i. Optimize Overall System Performance

Enhance the coordination and efficiency of all components within the Mini Plant to ensure smooth, reliable, and streamlined operations.

ii. Integrate Advanced PLC Technology

Implement state-of-the-art Programmable Logic Controllers (PLCs) to automate control processes, improve accuracy, and reduce manual intervention.

iii. Enable Centralized Monitoring through SCADA

Deploy a Supervisory Control and Data Acquisition (SCADA) system to provide centralized oversight of all plant activities, enabling real-time monitoring and improved management.

iv. Facilitate Localized Control with HMI

Install Human-Machine Interface (HMI) panels at key locations within the Mini Plant to allow operators direct control and interaction with specific system components.

v. Enhance Decision-Making with Real-Time Insights

Leverage SCADA-driven data to deliver up-to-date system performance metrics, supporting faster and more informed decision-making by operators and engineers.

2.3 Scope of Work

This Scope of Work defines the key tasks and responsibilities involved in the execution of the Automated Bottling Station Project, focusing on enhancing automation, monitoring, and overall system performance through the integration of advanced industrial technologies.

i. System Assessment

Perform a comprehensive evaluation of the existing Mini Plant setup to identify inefficiencies, potential upgrades, and areas requiring functional improvement.

ii. SCADA Implementation

Deploy a Supervisory Control and Data Acquisition (SCADA) system to enable centralized monitoring and control. Specific tasks include:

- Monitoring water levels in the tank.
- Controlling the on/off operation of motors remotely.

iii. HMI Installation

Install Human-Machine Interface (HMI) panels at strategic locations within the plant to facilitate localized control and provide operators with a user-friendly interface for real-time interaction with equipment.

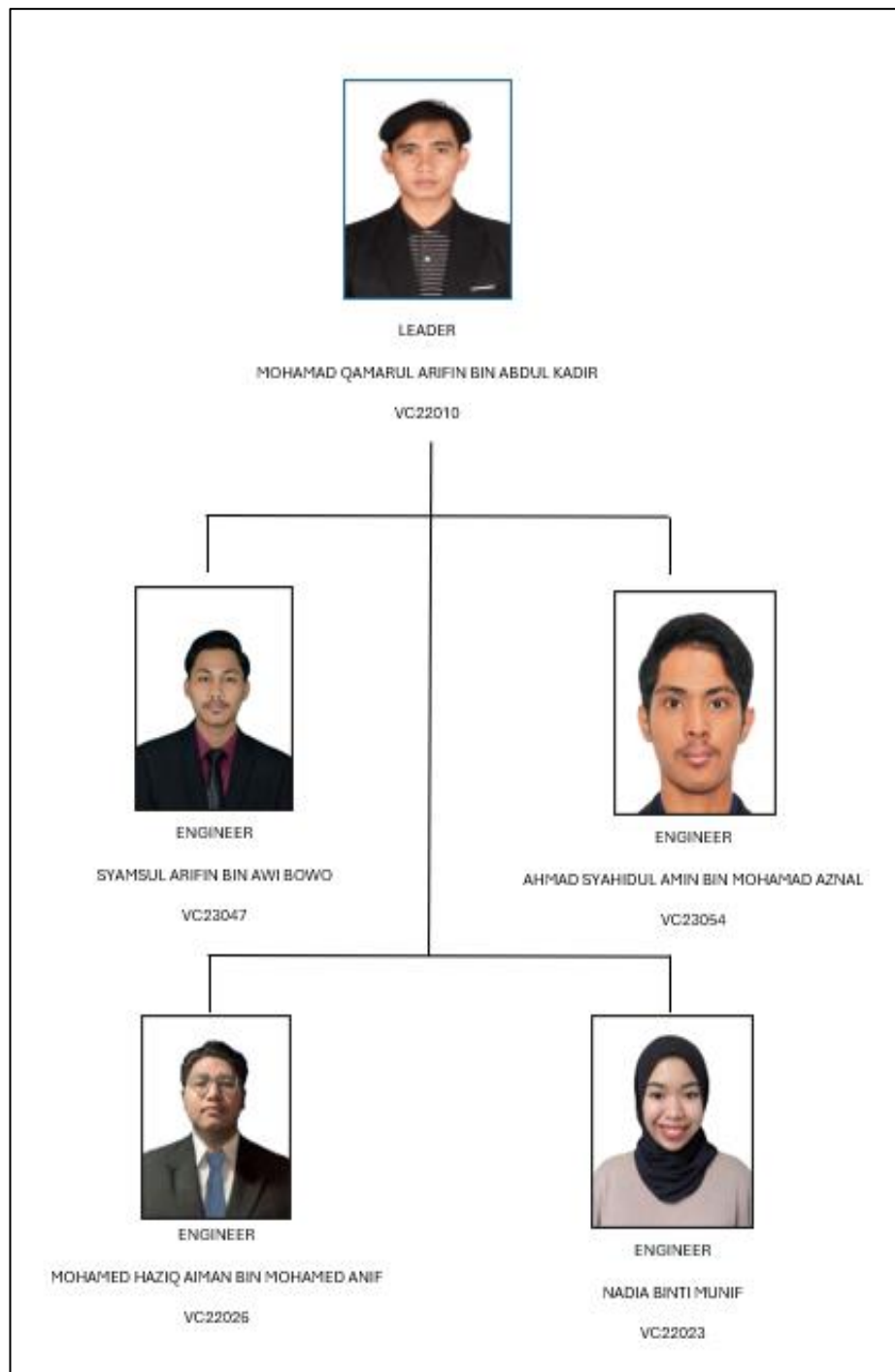
iv. Communication Protocols

Develop and configure robust communication protocols to ensure reliable and seamless data exchange between PLCs, SCADA systems, and HMI panels.

v. Performance Monitoring and Optimization

Implement systems for real-time performance tracking, data analysis, and continuous optimization to maintain high efficiency and adapt to evolving operational needs.

2.4 Organization Chart



Organization Chart SCADA and HMI Team

2.5 SCADA And HMI Team Task

NO.	NAME	TASK
1	 LEADER MOHAMAD QAMARUL ARIFIN BIN ABDUL KADIR VC22010	• Installation and setup for SCADA • Design a system for SCADA • Alarm handling and acknowledgement • Logging in and user access control
2	 ENGINEER SYAMSUL ARIFIN BIN AWI BOWO VC23047	• Installation and setup for HMI • Design a system for HMI • Complete all tasks related to HMI • Logging in and user access control
3	 ENGINEER AHMAD SYAHIDUL AMIN BIN MOHAMAD AZNAL VC23054	• Installation and setup for HMI • Design a system for HMI • Complete all tasks related to HMI • Logging in and user access control
4	 ENGINEER MOHAMED HAZIQ AIMAN	• Design a system for SCADA • Prepare project reports for the SCADA/HMI team. • Data visualization and reporting • Complete all tasks related to SCADA

	BIN MOHAMED ANIF VC22026	
5	 ENGINEER NADIA BINTI MUNIF VC22023	• Design a system for SCADA • Complete logbook • Data visualization and reporting • Prepare project reports for the SCADA/HMI team.

3.0 PROJECT DESCRIPTION

The Smart Mini Bottling System is an industrial automation project aimed at improving the efficiency, accuracy, and monitoring capabilities of a small-scale bottling process. The system is designed using a three-tier automation structure: Field Level, Control Level, and Supervisory Level.

Secondly, Field Level, various sensors and actuators are used to detect bottle presence, monitor water levels, control the flow of liquid, and operate motors and conveyors. These components enable precise control and real-time data collection from the physical system.

Thirdly, Control Level is managed by a Programmable Logic Controller (PLC), which automates and sequences operations such as bottle detection, filling, capping, and transferring. This level ensures smooth coordination among components, minimizes manual intervention, and improves process reliability.

Other than that, Supervisory Level, a combination of SCADA (Supervisory Control and Data Acquisition) and HMI (Human-Machine Interface) technologies are implemented to provide both centralized and localized control. SCADA allows operators to monitor the entire system remotely with real-time data visualization, while HMI panels placed near machinery offer user-friendly interfaces for local control and diagnostics.

The conclusion project aims to optimize bottling operations, reduce downtime, enhance production accuracy, and support decision-making through real-time insights. By integrating modern automation technologies, the Smart Mini Bottling System serves as a model for efficient, intelligent manufacturing and prepares students for real-world industrial applications.

In conclusion, the Smart Mini Bottling System enhances automation, accuracy, and monitoring through integrated PLC, SCADA, and HMI technologies—making it a reliable model for modern, efficient, and intelligent bottling operations.

4.0 SYSTEM ARCHITECTURE

4.1 Gantt Chart SCADA/HMI Team

No.	Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
1	Software Installation and Setup for SCADA/HMI								
2	Configuration from SCADA/HMI to PLC								
3	Graphic Design for SCADA/HMI								
4	Hand Over								
5	Redo Punchlist								
6	Get a tag management from PLC								
7	Design Face Plate								
8	Final Handover								
9	Redo Punchlist								

Table 1 : Gantt Chart

4.2 Automated Bottling Station System Flow

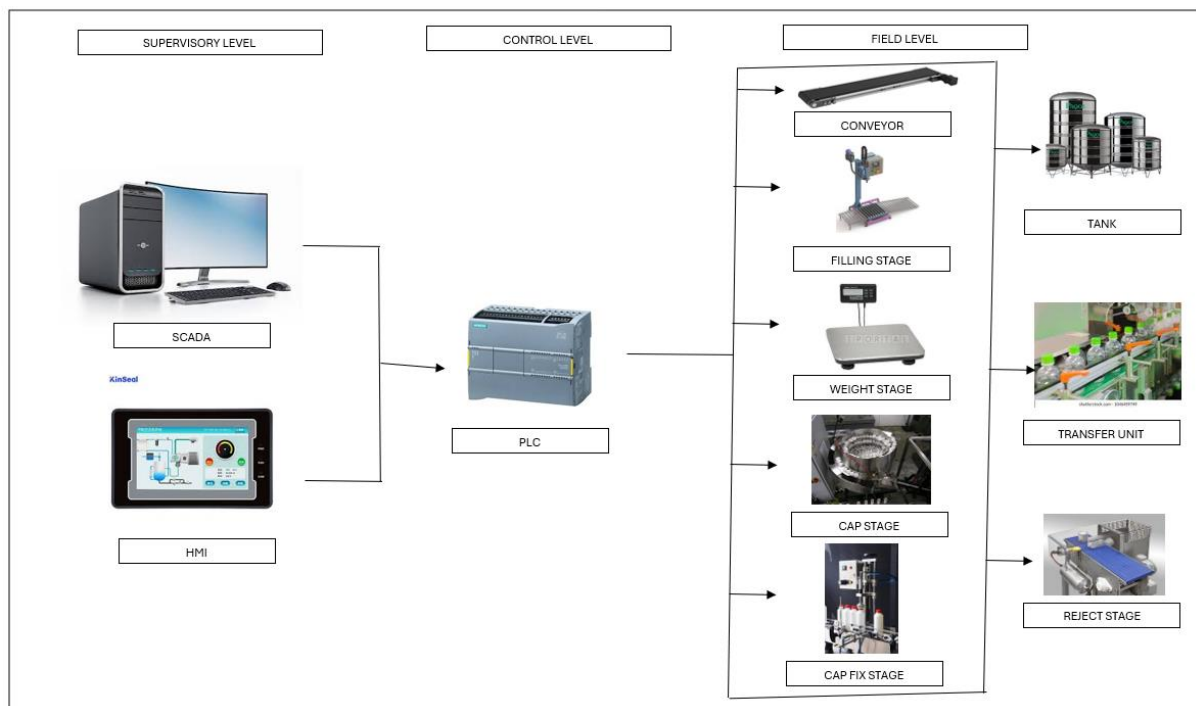


Figure 1 : Automated Bottling Station System

4.3 Flowchart Proses

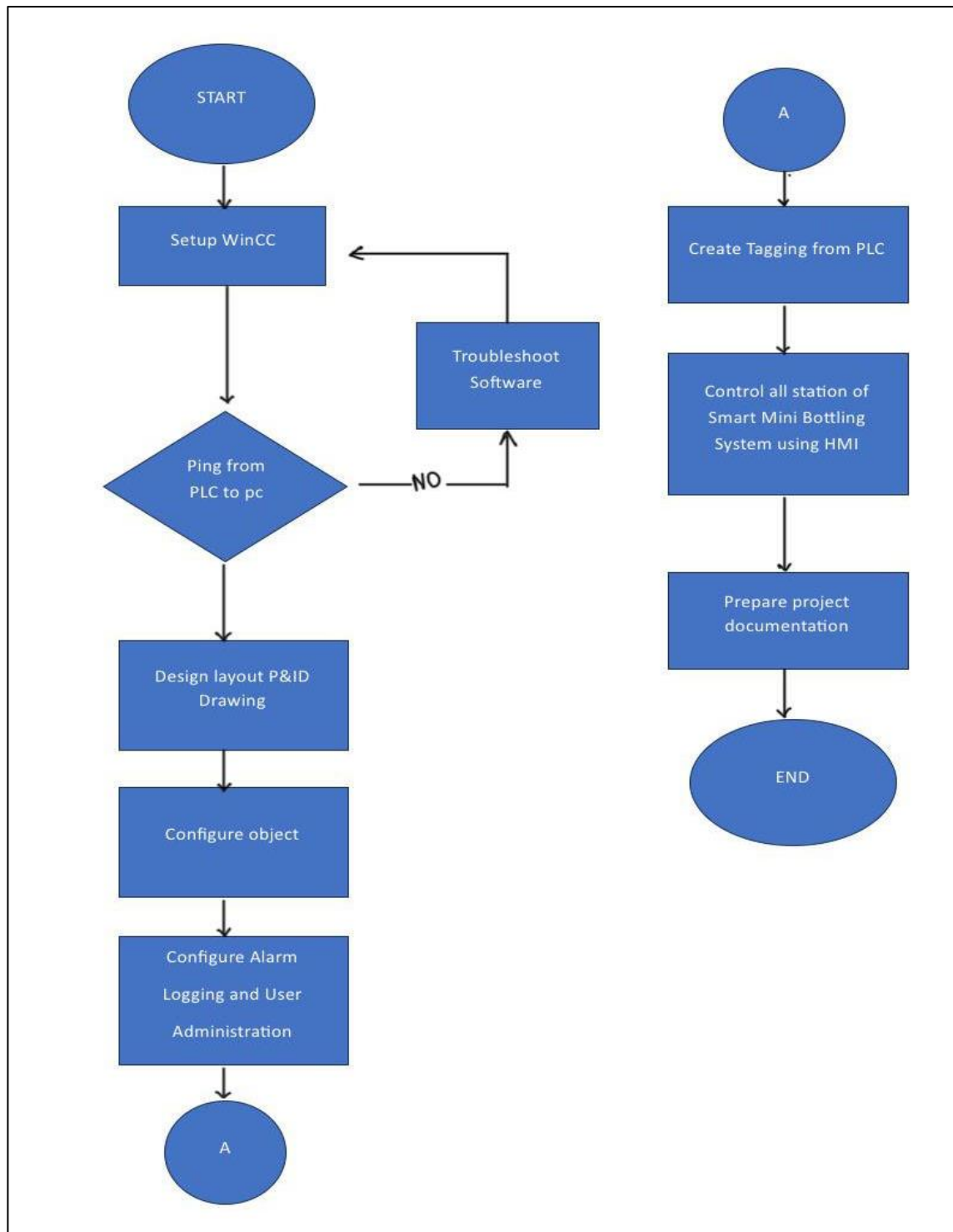


Figure 2 : Flowchart Proses

5. HARDWARE AND SOFTWARE SETUP

5.1 Scada Hardware & Software Setup

To begin a project in WinCC, open the WinCC Explorer and select "New" from the File menu. In the dialog that appears, you can either create a new project by selecting a project type or open an existing one. The next time you launch WinCC, it will automatically open the last project you worked on. If the project was active when you closed WinCC, it will reopen in the same state.

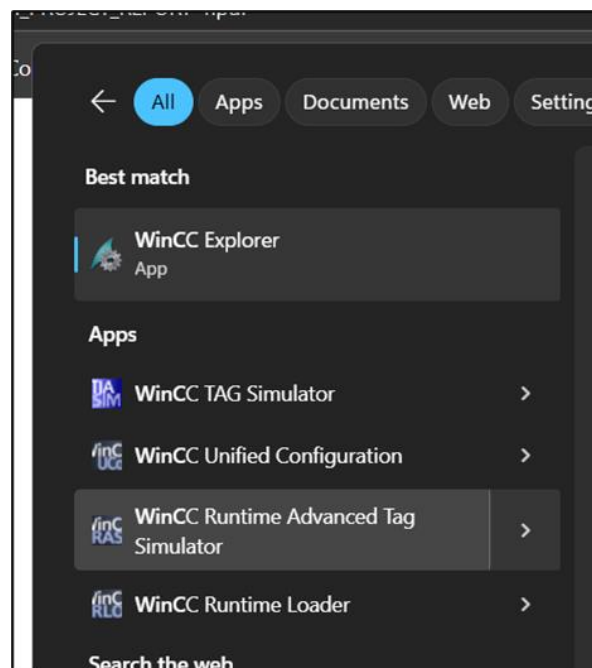


Figure 3 : Wincc Explorer Software

ii. Select project type "Single-user project".

The "Create new project" dialog box opens.

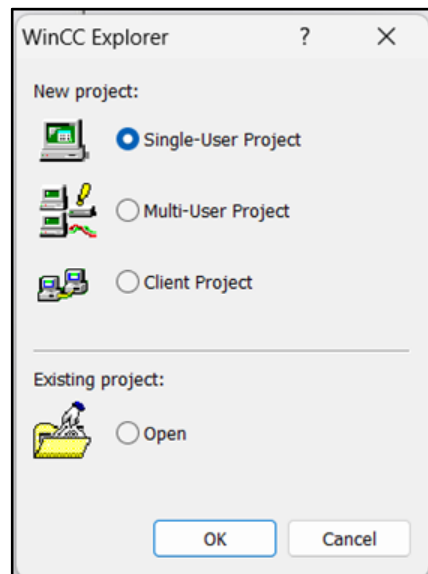


Figure 4 : New project in Wincc Explorer

iii. Enter the project information.

If you do not make changes in the fields "New subdirectory" and "Project path", the standard settings will be adopted.

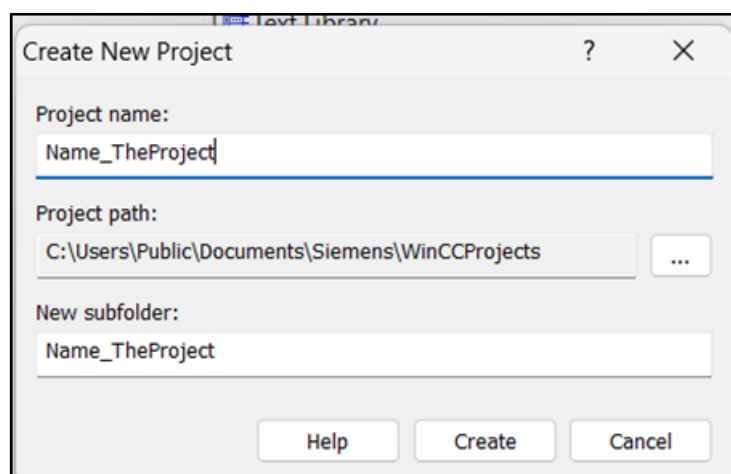


Figure 5 : Create New Project

You have created the project. The project is opened in the WinCC Explorer. The project structure with the necessary editors and directories is displayed in the left pane. of WinCC Explorer. The right-hand pane shows the elements that belong to an editor or directory.

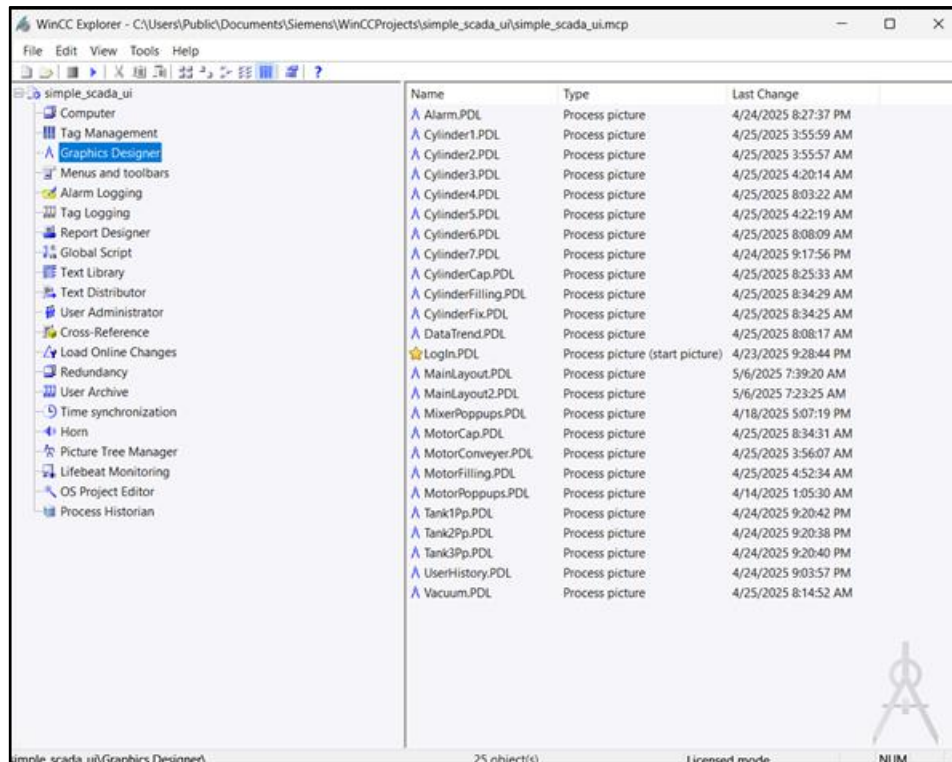


Figure 6 : Create layout for graphic designer

iv. **Pinging from SCADA PC to PLC**

Pinging from a SCADA system to PLCs involves using the command-line interface or a dedicated tool to send a series of network messages to the respective IP addresses of the PLCs. This process helps to verify the connectivity and responsiveness of the PLCs within the network. Here's a step-by-step guide on how to ping from SCADA to PLCs with the given IP addresses:

- a) Access Command Prompt or Terminal:
 - On a Windows-based SCADA system, open the Command Prompt.
 - On a Linux-based SCADA system, open the Terminal.
- b) Use the Ping Command:
 - Type the following command for PLC1: ping 192.168.0.10
 - Type the following command for PLC2: ping 192.168.0.1
 - Press Enter
- c) Observe Results:
 - The system will start sending ICMP (Internet Control Message Protocol) packets to the specified IP addresses. You will see a series of responses

indicating the time taken for each packet to travel from the SCADA system to the PLC and back. A successful ping will show responses like "Reply from [IP address]: bytes=32 time=[time in milliseconds] TTL=[time to live]." Unsuccessful pings may show "Request timed out" or "Destination host unreachable" messages.

d) Interpret Results:

- A successful ping indicates that the SCADA system can communicate with the PLC at the specified IP address. If pings are unsuccessful, it may suggest a network connectivity issue or misconfiguration.

e) Troubleshoot if Necessary:

- If there are issues with the ping, check the network configuration, including IP addresses, subnet masks, and gateways on both the SCADA system and the PLCs. Keep in mind that successful pinging establishes the foundation for communication between the SCADA system and the PLCs. This is crucial for real-time monitoring and control in industrial automation systems.

v. Tag management for PLC1 and PLC2

The tag addresses for PLC1 and PLC2 are displayed in the tag management system diagram below. These tag addresses are manually entered into the tag management system and correspond to their respective PLC addresses. Tags are essential in industrial automation, serving as unique identifiers for data points within the control system. Efficient tag management ensures precise communication between devices and PLCs, which is critical for seamless data exchange. This communication plays a major role in enabling the interface between PLCs and higher-level systems, such as Supervisory Control and Data Acquisition (SCADA) or Human Machine Interface (HMI).

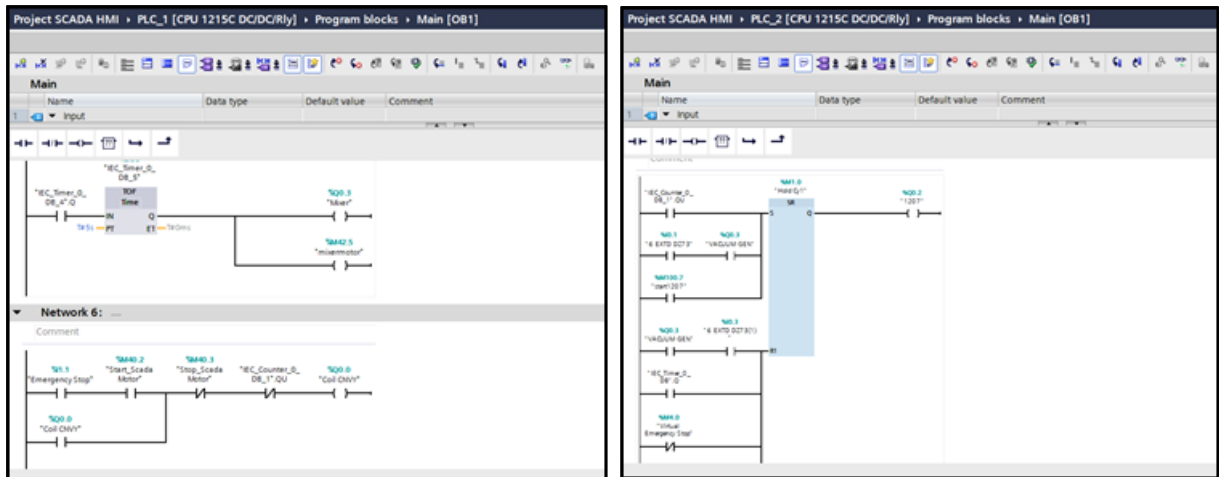


Figure 7 : PLC Ladder diagram

Tag Management - WinCC Configuration Studio

File Edit View Tools Help

Tag Management

Internal tags

SMATIC ST-1200, ST-1500 Channel

PLC

Structure tags

Actuator

Color

Motor

Tank

Weight

Tags [PLC]

Name	Data type	Length	Format adaptation	Connection	Group	Address	Linear scaling	AS value
1 Cylinder1Running(1200)Cov	Binary Tag	1		PLC		M40.6		
2 Cylinder2Running(1203)TULeft	Binary Tag	1		PLC		M41.2		
3 Cylinder3Running(1201)Cov	Binary Tag	1		PLC		M41.0		
4 Cylinder4Running(1204)TUFront	Binary Tag	1		PLC		M41.5		
5 Cylinder5Running(1202)Filling	Binary Tag	1		PLC		M41.7		
6 CylinderExpand1Start	Binary Tag	1		PLC		M40.7		
7 CylinderExpand2Start	Binary Tag	1		PLC		M41.1		
8 CylinderExpand3Start1203Left	Binary Tag	1		PLC		M41.3		
9 CylinderExpand4Start1204Front	Binary Tag	1		PLC		M41.4		
10 CylinderExpand5Start1202Fill	Binary Tag	1		PLC		M41.6		
11 DummyWaterFill	Unsigned 32-bit value	4	DwordToUnsignedDword	PLC				
12 FlowmeterIndicator1	Binary Tag	1		PLC		M42.3		
13 FlowmeterIndicator2	Binary Tag	1		PLC		M42.4		
14 GoodProductCountW	Binary Tag	1		PLC		M1.5		
15 MixerMotor	Binary Tag	1		PLC		M42.5		
16 MotorFillingRunning	Binary Tag	1		PLC		M42.2		
17 PhotoDetect1	Binary Tag	1		PLC		M40.4		
18 PhotoDetect2	Binary Tag	1		PLC		M40.5		
19 RealValueWeight	Floating-point number 32-bit IEEE	4	FloatToFloat	PLC		MD104		
20 RejectProductCountW	Binary Tag	1		PLC		M11.1		
21 RunningMotorCov	Binary Tag	1		PLC		M40.1		
22 StartMotorCov	Binary Tag	1		PLC		M40.2		
23 StartMotorFilling	Binary Tag	1		PLC		M42.0		
24 StopMotorCov	Binary Tag	1		PLC		M40.3		
25 StopMotorFilling	Binary Tag	1		PLC		M42.1		

Tag Management

Alarm logging

Tag Logging

Text Library

User Administrator

Ready NUM English (United States) Table: 25 Tags 100 %

Figure 8 : Tag Address PLC 1

Name	Data type	Length	Format adaptation	Connection	Group	Address	Linear scaling	AS value
1 ApprovedProductCount	Binary Tag	1		PLC2		M110.0	☐	
2 Cylinder6Running(1206)Cap	Binary Tag	1		PLC2		M101.6	☐	
3 Cylinder7Running(1207)Cap	Binary Tag	1		PLC2		M110.4	☐	
4 Cylinder8Running(1208)Fix	Binary Tag	1		PLC2		M102.2	☐	
5 Cylinder8Running(1209)PushReject	Binary Tag	1		PLC2		M102.6	☐	
6 Cylinder8Running(1313)GateReject	Binary Tag	1		PLC2		M102.5	☐	
7 CylinderExpand9PushReject	Binary Tag	1		PLC2		M103.0	☐	
8 CylinderExpand10GateReject	Binary Tag	1		PLC2		M102.7	☐	
9 EmergencyPB	Binary Tag	1		PLC2			☐	
10 OnMotorCap	Binary Tag	1		PLC2		M102.3	☐	
11 RejectProductCount	Binary Tag	1		PLC2		M110.1	☐	
12 RunMotorCap	Binary Tag	1		PLC2		M102.4	☐	
13 runningvacuum	Binary Tag	1		PLC2		M101.4	☐	
14 startCylinderExpand6_Cap1207	Binary Tag	1		PLC2		M100.7	☐	
15 StartCylinderExpand7_Cap1206	Binary Tag	1		PLC2		M101.5	☐	
16 StartCylinderExpand8_Cap1208	Binary Tag	1		PLC2		M102.0	☐	
17 startvacuum	Binary Tag	1		PLC2		M101.2	☐	
18 stopCylinderExpand6_Cap1207	Binary Tag	1		PLC2		M101.1	☐	
19 StopCylinderExpand7_Cap1206	Binary Tag	1		PLC2		M101.7	☐	
20 StopCylinderExpand8_Cap1208	Binary Tag	1		PLC2		M102.1	☐	
21 stopvacuum	Binary Tag	1		PLC2		M101.3	☐	
22								
23								
24								
25								
26								
27								
28								
29								
30								
31								
32								
33								
34								
35								

Figure 9 : Tag Address PLC 2

5.2 HMI HARDWARE & SOFTWARE SETUP

5.1.1 Configure Hmi

i. Create a new project

Navigate to TIA PORTAL and create a new project. Then fill in all the necessary information about the project.

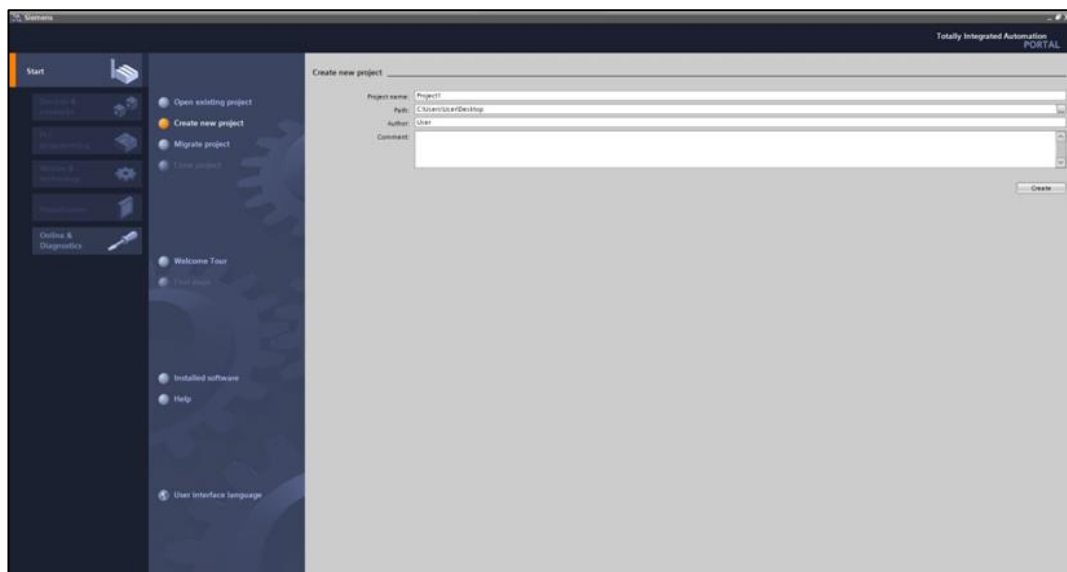


Figure 10 : Create New Project

ii. Add device

Scroll through devices & networks and find/click “Add new device”

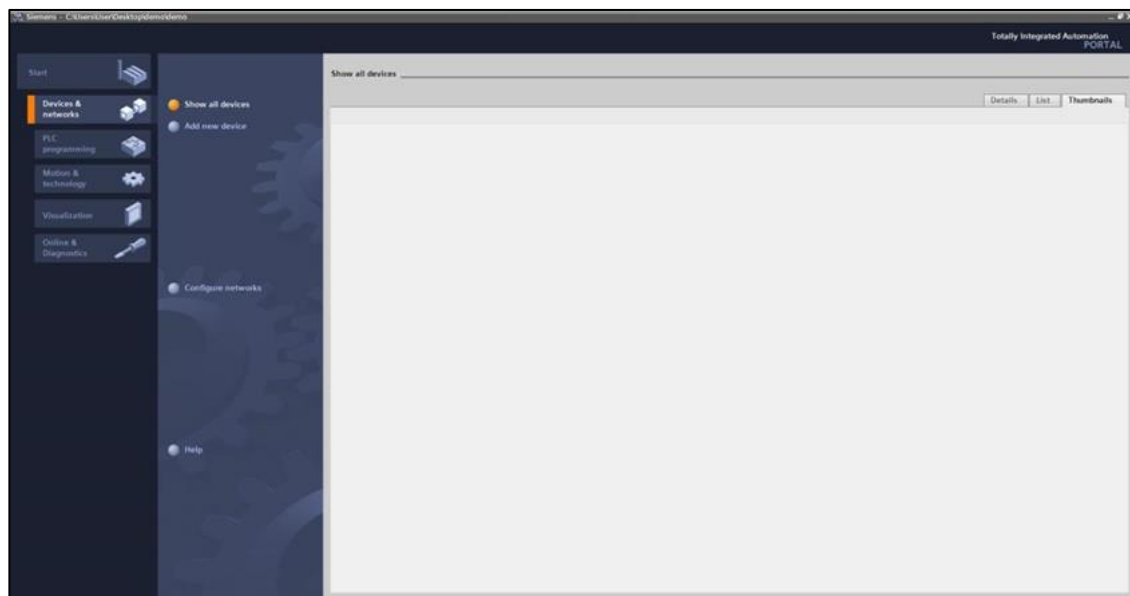


Figure 11 : Add new devices

iii. Add HMI

Go through HMI/Simatic Basic Panel/7” Display/KTP700 Basic/6AV2 1232GB03-0AX0 and select version 15.1.0.0

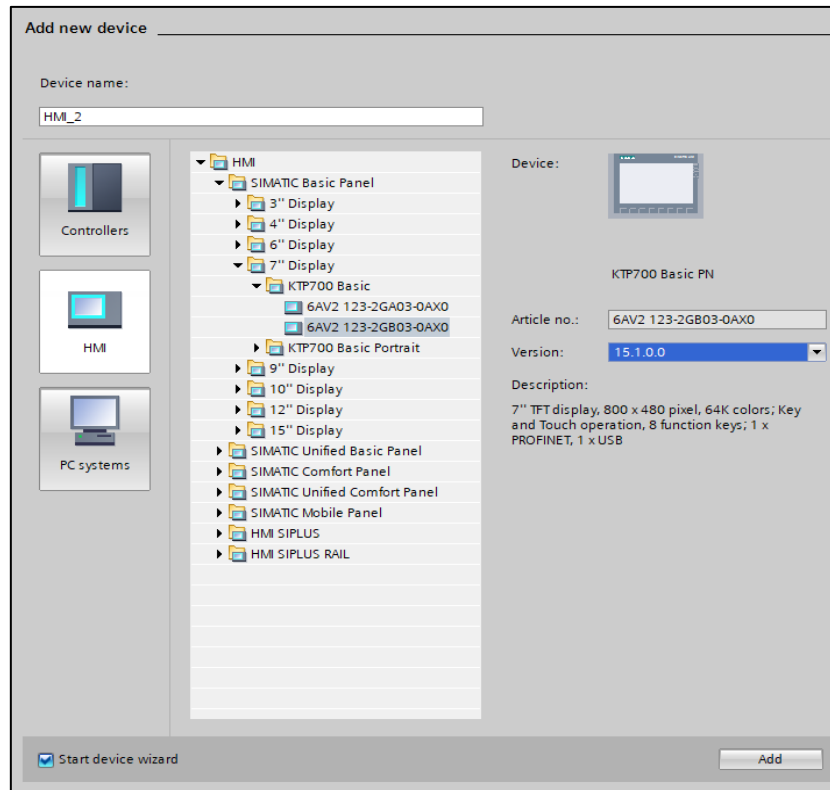


Figure 12 : Add HMI in TIA Portal

iv. Add PLC

Select the PLC for this project and start to configure your HMI display settings such as alarm and screen setup or just simply click finish.

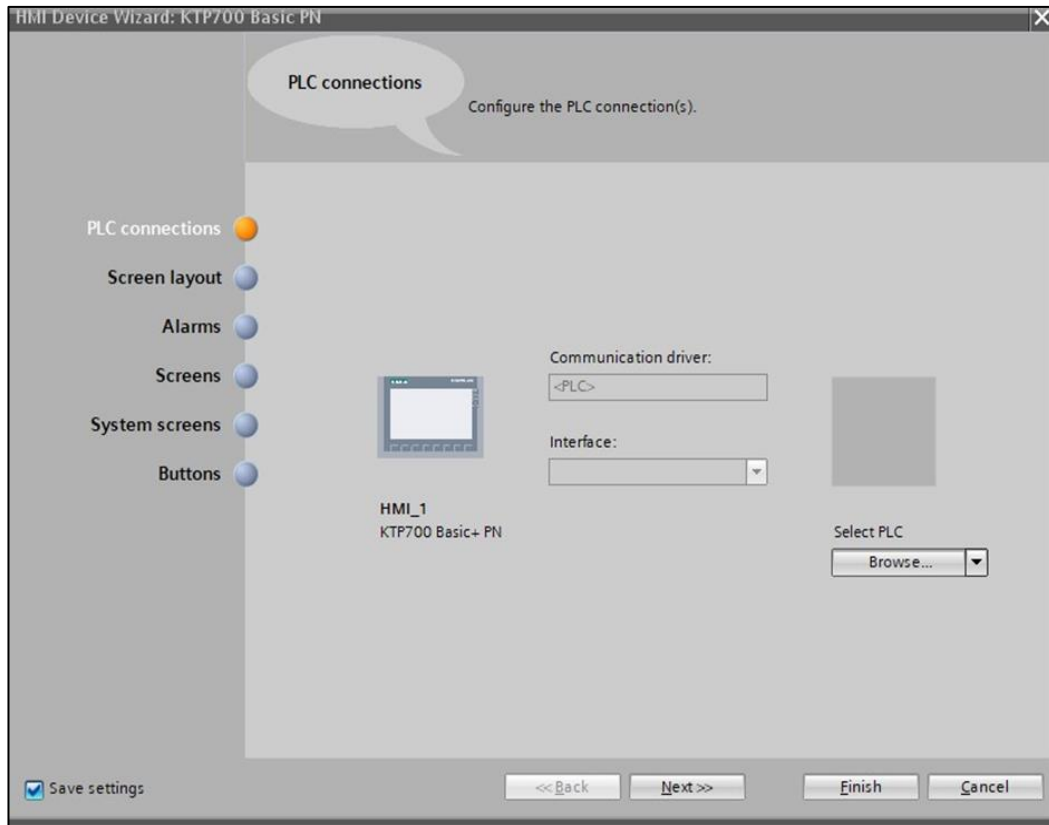


Figure 13 : Add PLC in TIA Portal

5.1.2 Screen Configuration

iv. Setting the start screen

Before adding the button, it's important to understand the concept of a start screen. The start screen is the first interface that appears when the HMI (Human Machine Interface) system begins operation. This screen can be set by selecting it explicitly or configuring it through the runtime settings.

In our current setup, a screen titled "Login Screen" is designated as the start screen. To make Screen One serve as the home screen, we can rename it to "Home Screen" and set it as the new start screen. This ensures that the Home Screen is the first screen displayed when the HMI is launched.

v. Adding a Button

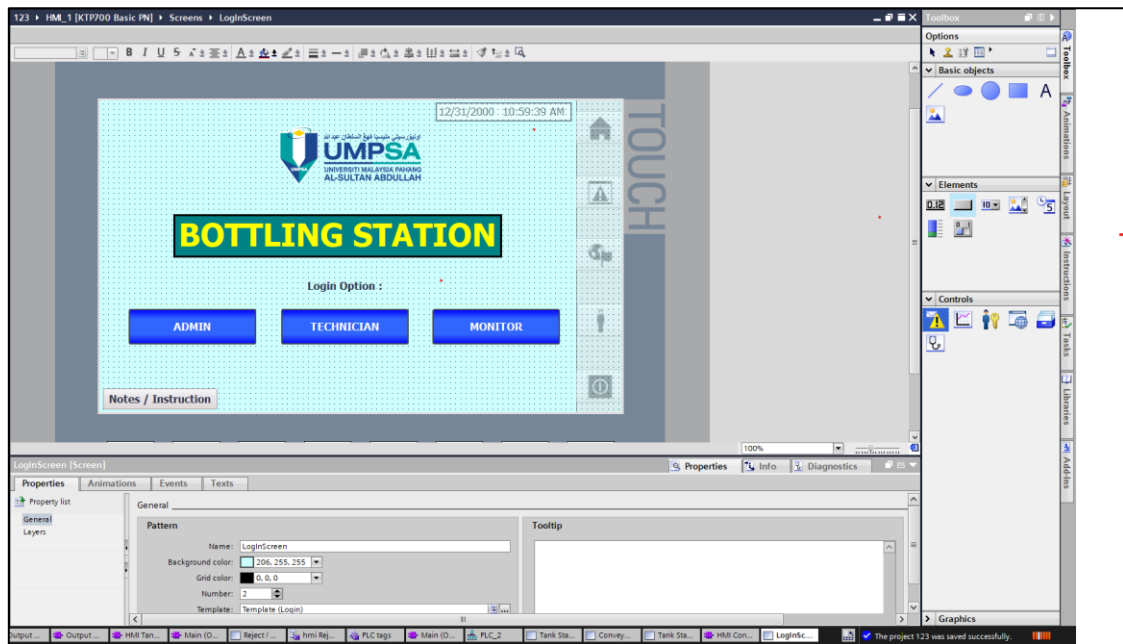


Figure 14 : Adding a Button

"First, open the toolbox on the right side in TIA Portal and drag a 'Button' element onto the HMI screen. Then, configure its properties as needed."

vi. Configuring the Button

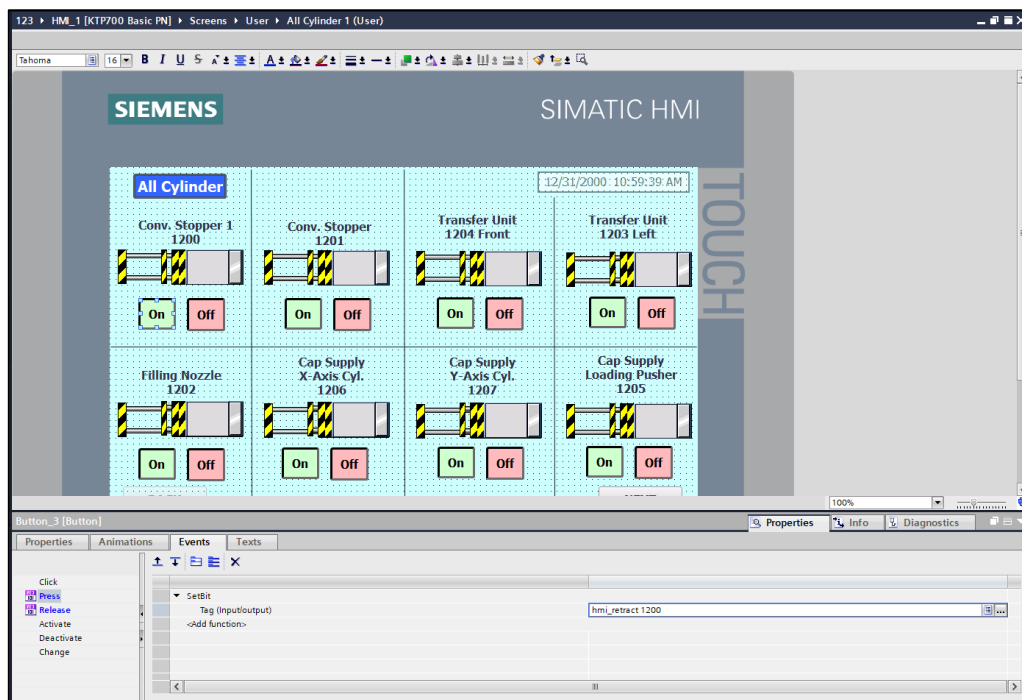


Figure 15 : Configuring the Button

In the Events section, select the 'Press' event, then add a function by choosing 'SetBit'. Next, assign the appropriate tag; it's an input tag.

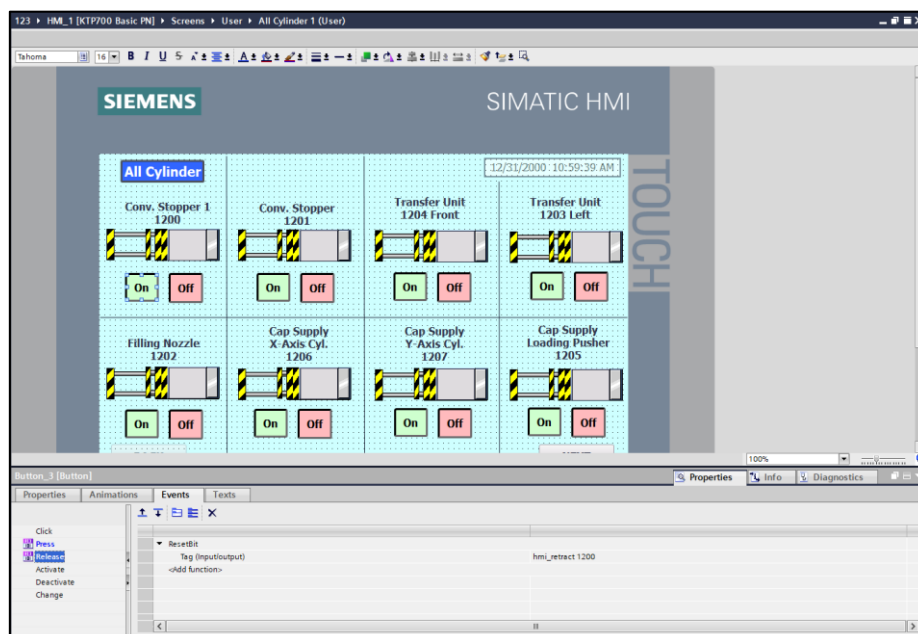


Figure 16 : ResetBit in HMI TIA Portal

"Just like the 'Press' event, select the 'Release' event, then add a function by choosing 'ResetBit'. After that, assign the appropriate tag (input)."

"For the Off button, follow the same configuration steps as you did for the On button (above)".

vii. Dynamic Colour Change

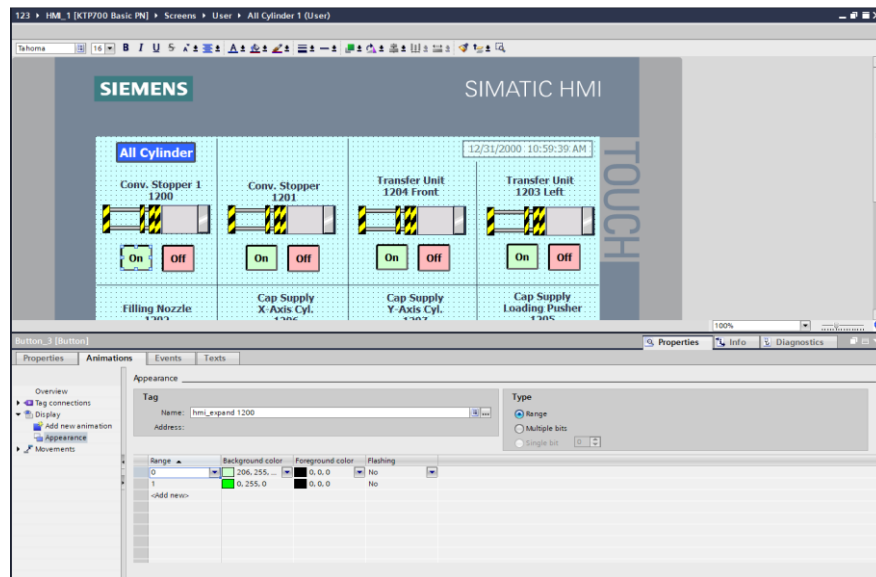


Figure 17 : Dynamic Colour Change in HMI

"In the Properties menu, go to the 'Animations' section, then select 'Display' and add a new animation. For the tag, bind it to your HMI tag. Set the colors for bit values 0 and 1 to indicate whether the signal is de-energized or energized."

Note: These steps are applicable to all HMI buttons you add in your project.

5.1.3 HMI Tags

i. HMI Tag (Input)

Stage Name	Description	Address	Remark
Home screen	hmi_start system	m30.0	PLC 1
	hmi_stop system	m30.1	
	hmi_emergency stop	m30.2	
	hmi_start system 2	m30.0	PLC 2
	hmi_stop system 2	m30.1	
	hmi_emergency stop 2	m30.2	
Tank stage	hmi_start mixer	m30.3	

	hmi_stop mixer	m30.4	
	hmi_start flowmeter 1	m30.5	
	hmi_start flowmeter 2	m30.6	
Conveyor stage	hmi_start conv. motor	m30.7	
	hmi_stop conv. motor	m31.0	
	hmi_expand 1200	m31.1	Conveyor Stopper 1
	hmi_retract 1200	m31.2	
	hmi_expand 1201	m31.3	Conveyor Stopper 2
	hmi_retract 1201		
Transfer unit stage	hmi_expand 1204	m31.5	Transfer Unit Front
	hmi_retract 1204	m31.6	
	hmi_expand 1203	m31.7	Transfer Unit Left
	hmi_retract 1203	m32.0	
Filling stage	hmi_start pump (hold)	m32.1	Hold for future
	hmi_stop pump (hold)	m32.2	
	hmi_open solenoid	m32.3	Filling Solenoid
	hmi_close solenoid	m32.4	
	hmi_expand 1202	m32.5	Filling Nozzle
	hmi_retract 1202	m32.6	
Weighing stage	RealWeight	md104	
Cap supply stage	hmi_expand 1206	m32.7	Cap Supply X- Axis
	hmi_retract 1206	m33.0	
	hmi_expand 1207	m33.1	Cap Supply Y- Axis
	hmi_retract 1207	m33.2	
	hmi_expand 1205	m33.3	Cap Loading
	hmi_retract 1205	m33.4	
	hmi_on vacuum gen	m33.5	
	hmi_off vacuum gen	m33.6	
Cap fix stage	hmi_start fix motor	m33.7	
	hmi_stop fix motor	m34.0	
	hmi_expand 1208	m34.1	Cap Fix Motor position
	hmi_retract 1208	m34.2	
reject / good stage	hmi_expand 1313	m34.5	Reject Gate

	hmi_retract 1313	m34.6	Reject Pusher
	hmi_expand 1209	m34.7	
	hmi_retract 1209	m35.0	

ii. HMI Tag (Output)

Stage Name	Description	Address	Remarks
Tank stage	hmi ind_mixer	m50.0	
	hmi ind_flowmeter 1	m50.1	
	hmi ind_flowmeter 2	m50.2	
Conveyor stage	hmi ind_conv. motor	m50.3	
	hmi ind_1200	m50.4	Conveyor Stopper 1
	hmi ind_1201	m50.5	Conveyor Stopper 2
Transfer unit stage	hmi ind_move front 1204	m50.6	Transfer Unit Front
	hmi ind_move left 1203	m50.7	Transfer Unit Left
Filling stage	hmi ind_pump (hold)	m51.0	Hold for future
	hmi ind_solenoid	m51.1	
	hmi ind_1202	m51.2	Filling Nozzle
Weighing stage	Approved	m1.4	Good Weight Bit Signal
	Reject	m11.1	Reject Weight Bit Signal
	RealWeight	md104	
Cap supply stage	hmi ind_1206	m51.5	Cyl X-Axis
	hm ind_1207	m51.6	Cyl Y-Axis
	hmi ind_1205	m51.7	Cyl Load Bottle Cap
	hmi ind_vacuum gen	m52.0	
Cap fix stage	hmi ind_fix motor	m52.1	
	hmi ind_1208	m52.2	Cyl Fix Motor Position
Reject / good stage	hmi ind_1313	m52.4	Reject Gate
	hmi ind_1209	m52.5	Reject Pusher
	hmi_reject count	mw30	
	hmi_good count	mw31	

All cylinders 1	hmi ind_1200	m50.4	Conveyor Stopper 1
	hmi ind_1201	m50.5	Conveyor Stopper 2
	hmi ind_move front 1204	m50.6	Transfer Unit Front
	hmi ind_move left 1203	m50.7	Transfer unit Left
	hmi ind_1202	m51.2	Filling Nozzle
	hmi ind_1206	m51.5	Cap Supply Cyl X-Axis
	hmi ind_1207	m51.6	Cap Supply Cyl Y-Axis
	hmi ind_1205	m51.7	Cap Supply Cyl Load Bottle Cap
All cylinders 2	hmi ind_1208	m52.2	Fix Motor Position
	hmi ind_1313	m52.4	Reject Gate
	hmi ind_1209	m52.5	Reject Pusher

5.1.4 Upload Program to HMI

i. Connect the HMI to Your PC

- Use an Ethernet connection to establish a physical link between your workstation and the KTP700 HMI. ii. Launch TIA Portal
- Open the TIA Portal software on your workstation

ii. Set IP Address

- Set both HMI their IP address and make a virtual connection in TIA portal
✓ HMI_1 :192.168.0.2
- Make sure to ping from computer to know both HMI has been connected and data received

```

C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time=13ms TTL=128
Reply from 192.168.0.2: bytes=32 time=5ms TTL=128
Reply from 192.168.0.2: bytes=32 time=8ms TTL=128
Reply from 192.168.0.2: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 13ms, Average = 8ms

```

Figure 18 : Ping to HMI

- Under "Transfer, Network & Internet", select the icon to navigate to the network settings.



Figure 19 : HMI Panel

- In the menu item "Interface PN X1", set the IP address under "IP address" and the subnet mask under "Subnet mask".



Figure 20 : HMI Menu Item

iii. Configure the Date and Time

- Under "System Setting", select the icon to navigate to the Date & Time settings.



Figure 21 : HMI System Setting

- In the menu item "Date & Time", set the Day, Month, Year, Hour and Minutes then the Time Shift should be left in default setting.

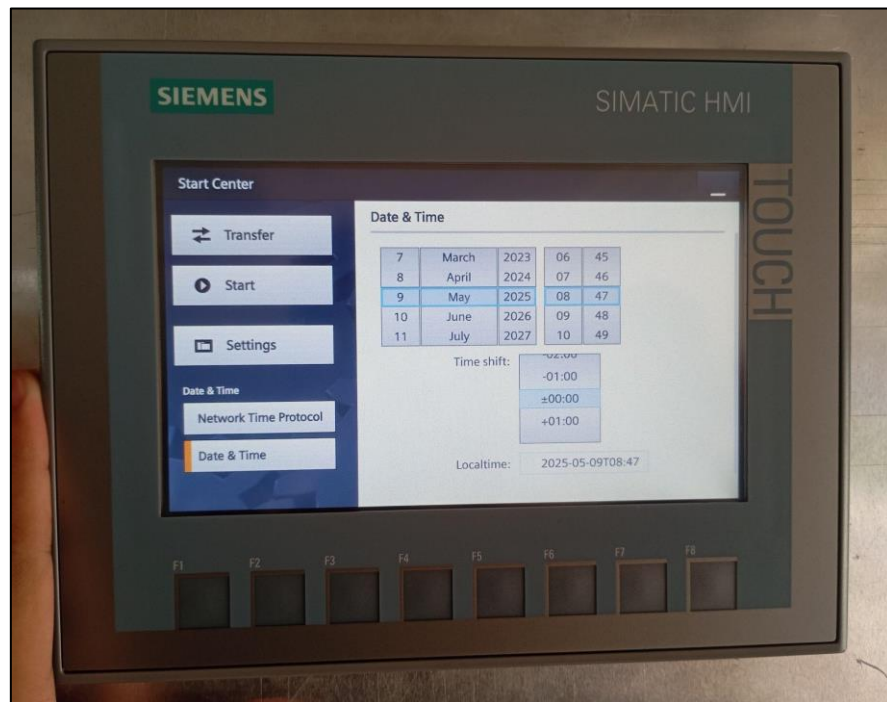


Figure 22 : HMI Date & Time

iv. Configure The HMI Connection

- Navigate to the project settings or hardware configuration within TIA Portal.
- Select the KTP700 HMI as the target device, and ensure that its IP address is set to **192.168.0.2**.

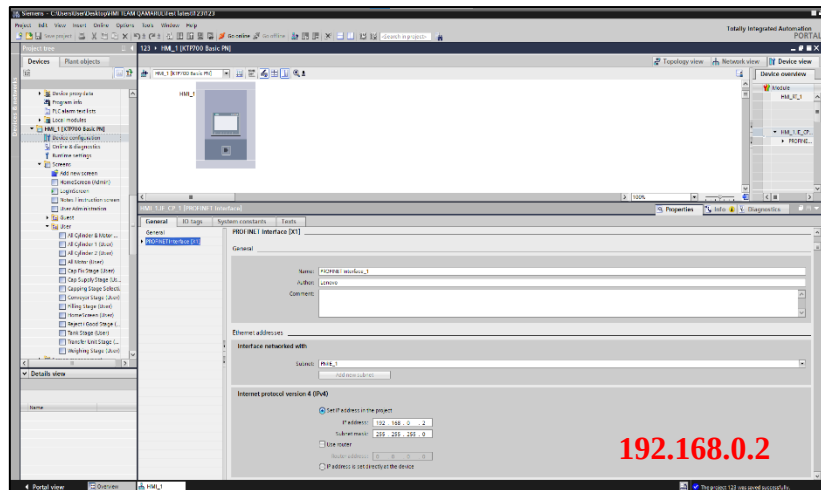


Figure 23 : Configure The HMI Connection

- Establish the connection between the HMI and PLC_1 in the HMI's device configuration.

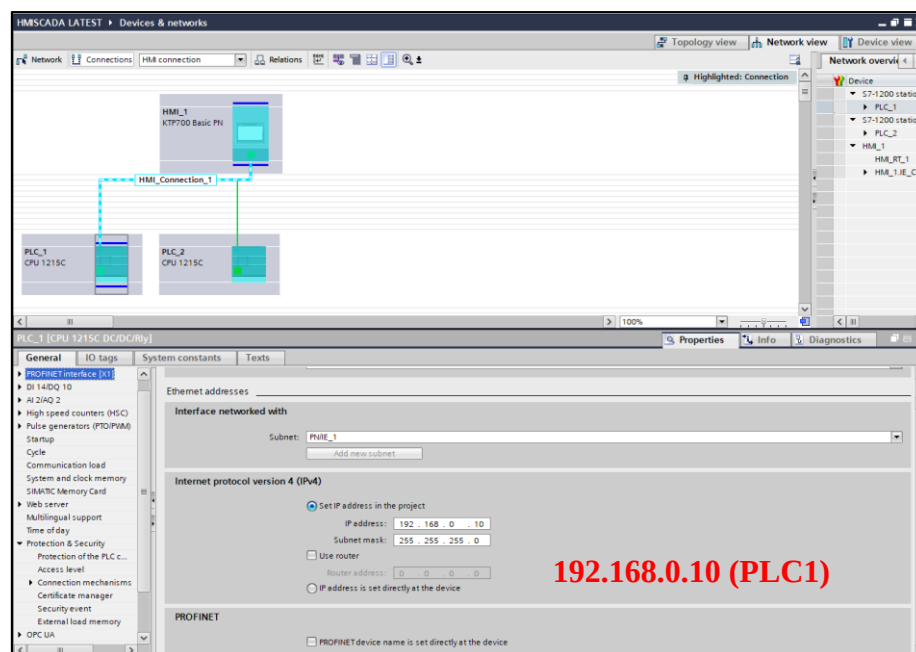


Figure 24 : 192.168.0.10 (PLC1)

- Establish the connection between the HMI and PLC_2 in the HMI's device configuration.

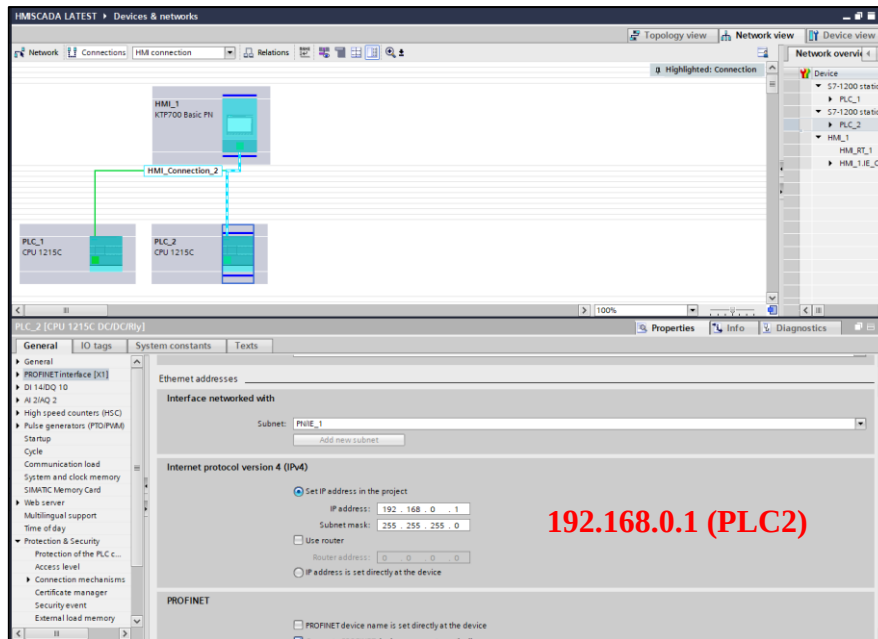


Figure 25 : 192.168.0.1 (PLC2)

v.Load the Project

- Upload the program to KTP700 HMI (download to devices).

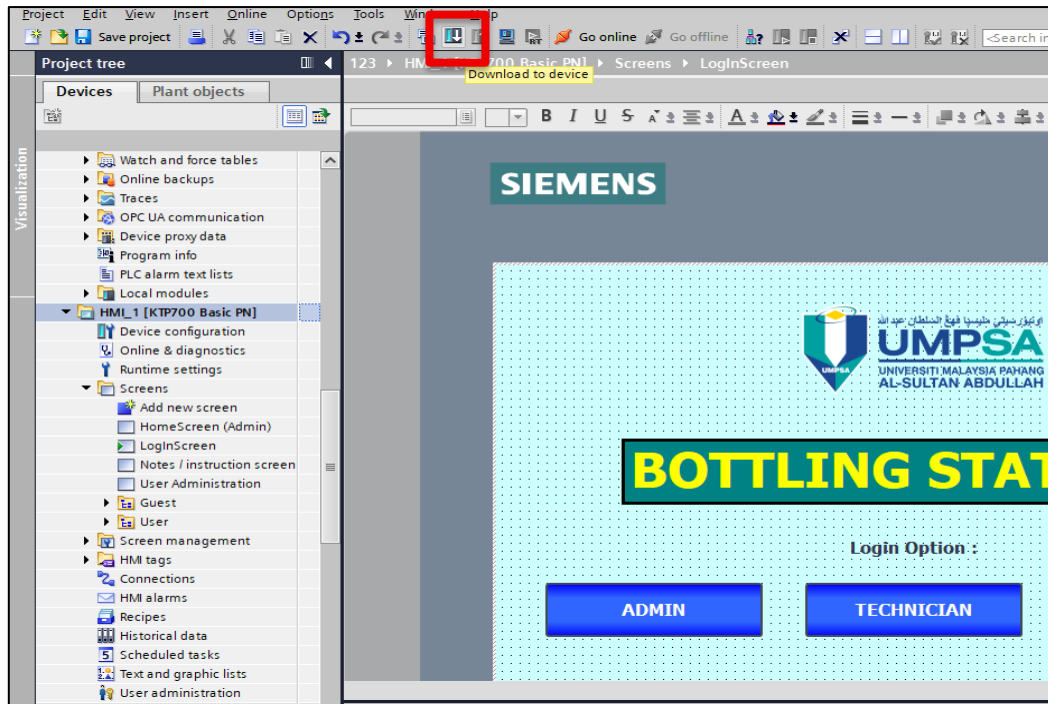


Figure 26 : Download Program to HMI

vi. Download the Project the HMI

- "Select the 'PG/PC Interface' connection, typically using Realtek (select the right device).

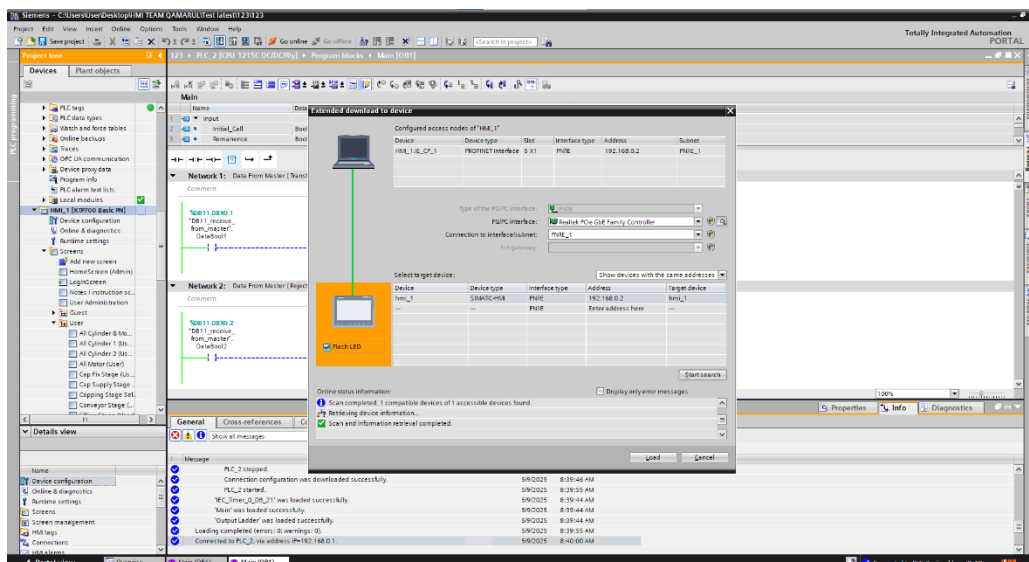


Figure 27 : Download Program to HMI (2)

- Before loading the HMI program, make sure to check the 'Overwrite All' option.

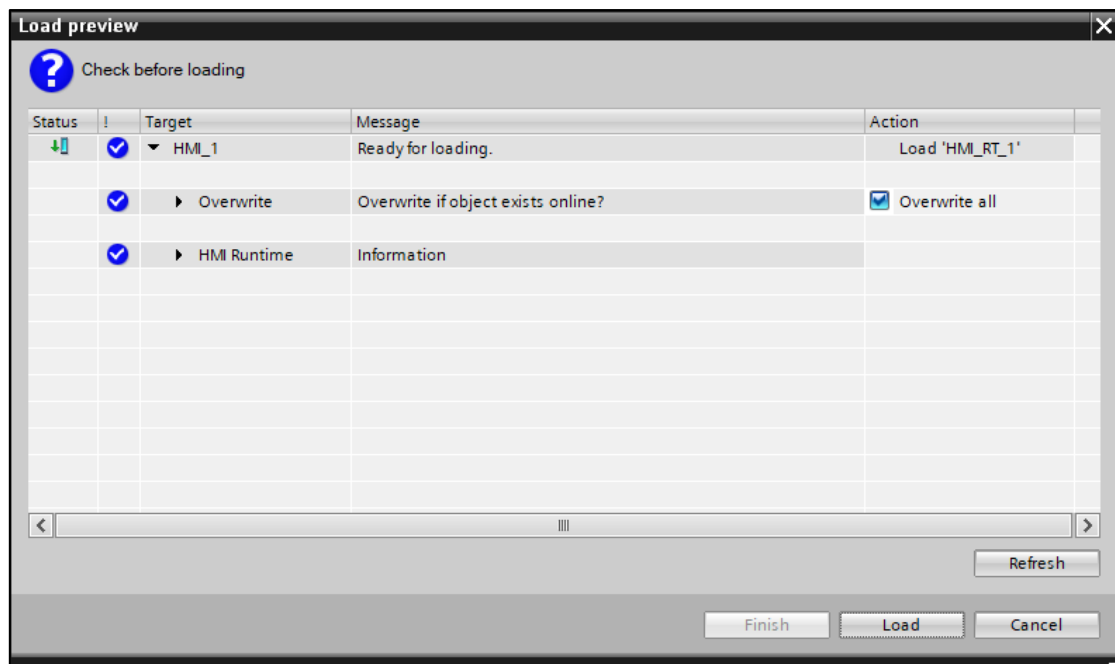


Figure 28 : Download Program to HMI (3)

vii. Review and Verify

- Wait for the process to complete.

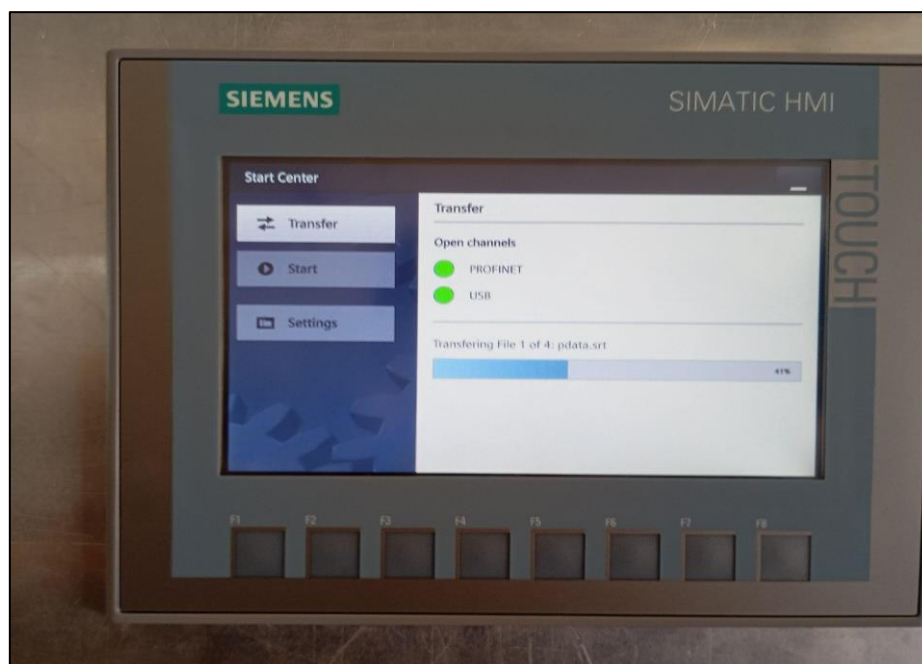


Figure 29 : Review and Verify

- Once done, the HMI screen will restart and an alarm showing connections are established, displaying the uploaded project in runtime.

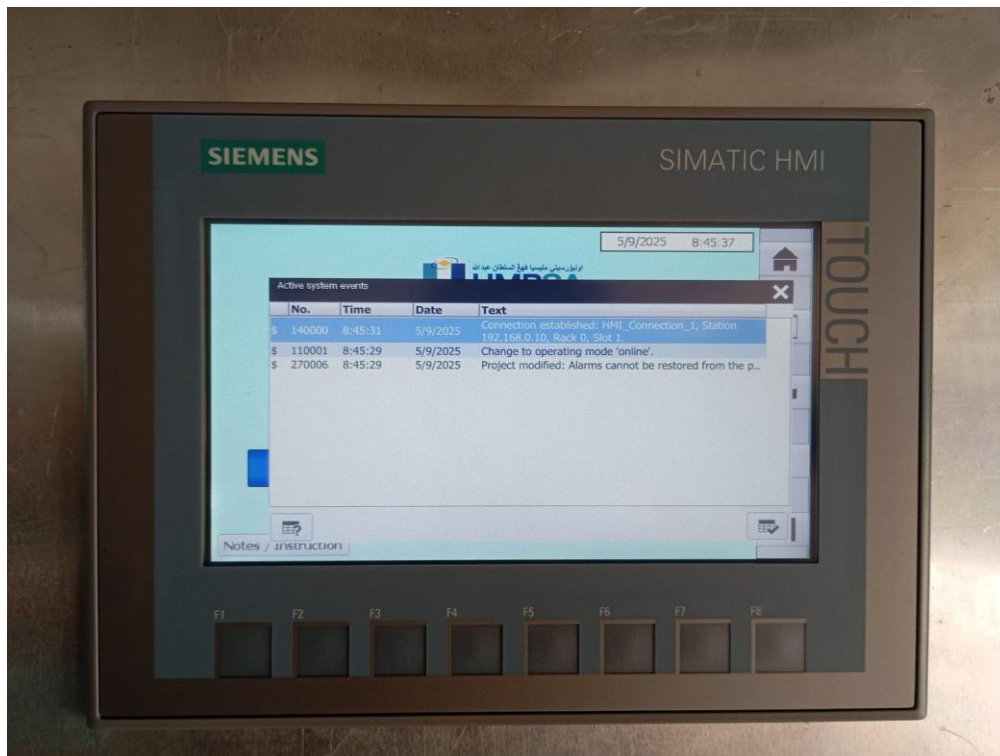


Figure 30 : Connection Establish HMI

Important Notes:

- Always maintain backups of your original project to prevent data loss during transfers.
- Refer to the specific user manuals or documentation provided by Siemens for detailed instructions tailored to your setup.
- Use proper cables and adhere to safety guidelines while connecting devices.

6.0 SCADA OPERATION

6.1.1 INSTALLATION AND SETUP WINCC 7.5

i. Introduction

This section describes how to install and run WinCC. The components already installed are displayed during setup. The following icons are used:

Icon	Meaning
	Current version of program is installed.
	Program is being updated.
	Program setup conditions are not met. Click the symbol for more detailed information.
	Program can be selected.
	Program selected for installation.
	Program cannot be selected (due to dependence on other programs).
	Program selected for installation (cannot be deselected).

ii. Scope of Installation

During custom installation of WinCC, you can choose between the following variants:

Typical	• WinCC Runtime • WinCC CS • Basic Process Control • SQL Server
Complete	"Typical", including: • OPC servers • SmartTools
Expert mode	Custom installation: You can select or deselect individual components in "WinCC Expert".
File server	• WinCC Fileserver • SQL Server
WinCC client ¹⁾	• WinCC Runtime • WinCC CS • Basic Process Control • SQL Express

Figure 31 : Installation scope

- If you are installing the "WinCC Client", you need an "RT Client" or "RC Client" client license.
- You can also install or remove components and languages at a later time. Read the sections "How to perform a supplementary installation" and "How to perform a supplementary installation of languages" for more on this.
- The required drive space depends upon the installed components. An estimated value is shown in the status bar. **Installation of WinCC Options**
- You can installed the desired options during the installation of WinCC itself. The documentation for some of the options will be available only if the concerned option package is installed.
- **Automatic Migration when a WinCC Project of a Previous Version is Opened**
When you open a project that was created with a version older than WinCC V7.4, the configuration data and Runtime data are automatically migrated. Convert the pictures and libraries with the Project Migrator or manually via the Graphics Designer. You can find detailed information about migration in the WinCC Information System under "First Steps > Migration".

iii.Requirements

- Make sure that no other setup is running on the computer at the same time, for example, a Windows update.
- You need local administrator rights to install WinCC.

Information on user rights, which is necessary for the operation of WinCC, is located in section "Instructions for Security of Data and System".

- The computer name may only contain permissible characters.
- The Windows component "MS Message Queuing" services must have been installed.
- The security policies must be adapted under Windows.
- No manually created SQL server entity with the name "WinCC" may be installed.
- The storage medium with the licenses is still not to be connected with the installation computer.
- If you want to use the OPC-XML-DA-Server from WinCC, the Microsoft Internet Information Service (IIS) must be installed before installing the OPC-XML-DA-Server.

WinCC is released for the following operating system languages: English, German, French, Italian, Spanish, Chinese simplified (PRC), Chinese traditional (Taiwan), Japanese, Korean and multi-lingual operating system.

Note:

An error message is output if you run WinCC Setup without having the administrator rights, or if other setup conditions are not met. You can find additional information on error messages under "WinCC Installation Requirements."

iv. Procedure

- a) Start the WinCC product DVD.

The DVD starts automatically if Autorun is enabled in the operating system. If the Autorun function is not activated, start the program Setup.exe on the DVD.

- b) Follow the on-screen instructions.

Read the License Agreement and the Open-Source License Agreement.

- c) Select the languages you want to install.

You may install other languages later.

- d) Select "Install" as the setup type.

If an older WinCC version is found, you can also activate the "Update" setup type. With this setup type, it is not possible to select additional products to be installed.

- e) Select the setup mode.

- f) In Package installation, select the Program package "WinCC Installation".

- If you also want to install WinCC options, select the corresponding program packages.
- Select "WinCC Client Installation" if you only want to install the WinCC client.
- Select the scope of your installation in User-defined installation.

The components to be installed are highlighted in Setup.

Click on "Help" for a description of the displayed symbols. Click on "Readme" to open the Information System.

- g) Read the license agreement for the Microsoft SQL Server.

- h) Before the installation, the security settings that are adapted for WinCC are displayed in the "System Settings" dialog. The firewall is configured automatically.

Confirm the changes to the system settings.

- i) Start the installation.

You can track the status of the installation in the displayed dialog. Select "Cancel" to cancel the installation of the current component.

- j) You can transfer the product License Keys after having installed the components.

To do so, click on "Transfer License Key". Select "Next" if you have already transferred the license keys or want to install them at a later time.

Note:

Transferring the licenses

The license keys will not be transferred automatically. You will have to transfer missing license keys during or after installation with Automation License Manager.

- k) Restart the computer to conclude the installation.

v. New entries in the "Siemens Automation" program group

After the installation of WinCC, you will find the new folders in the "Siemens Automation" program group.

- a) Starting WinCC Explorer:

- WinCC Explore

- b) Tools for working with WinCC:

- Channel Diagnosis
- Cross Reference Assistant
- Dynamic Wizard Editor
- Project Duplicator
- Project Migrator
- Tag Export Import
- WinCC Documentation Viewer
- WinCC Tag Simulator

- l) Documentation for WinCC in the WinCC Information System:

- WinCC Information System in the language in which WinCC was installed:
 - ✓ WinCC Information System.
- Documentation in all languages: Documentation • Print versions of the WinCC Information System:
 - ✓ PDF files in the installation path under "WinCC > Documents"

m) Management of the licenses:

- Automation License Manager.
- License Analysis

n) Security Controller for display of the customized security settings:

- Security Controller

o) Overview of the installed SIMATIC software and the components:

- Inst. Software

6.1.2 LOGGING IN AND USER ACCESS CONTROL

Navigate to user administrator to control or edit on how an admin will control the system.

Here you can set the username and password for the system.

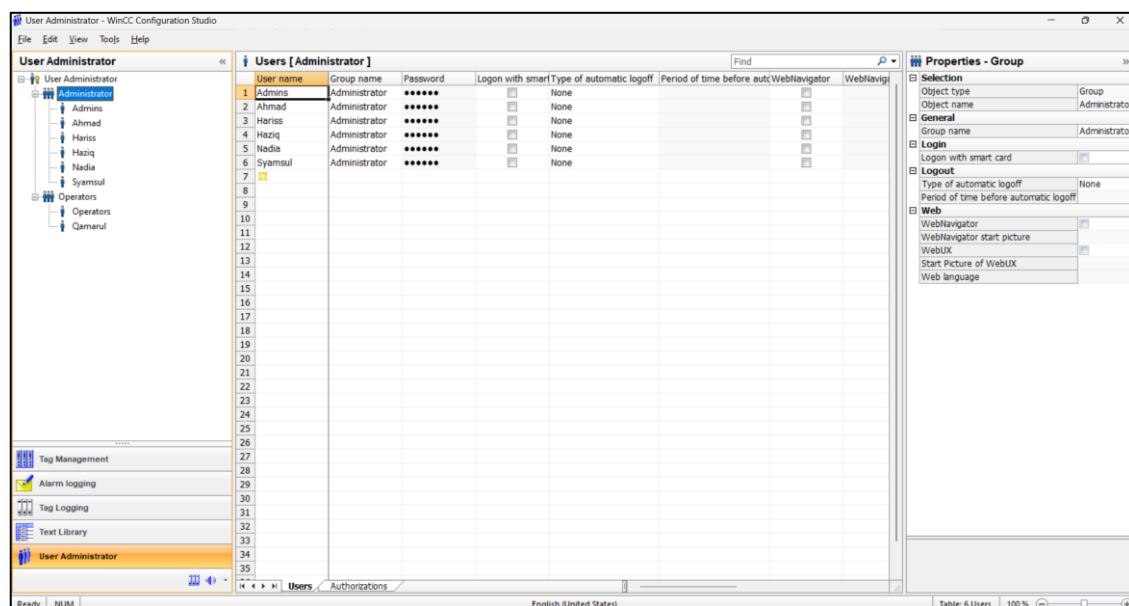


Figure 32 : User administrator

As an administrator, you can define user authorizations to control access within the system. This allows you to specify which users are permitted to perform certain actions or

access specific features. By assigning appropriate permissions, you help ensure that only authorized personnel can interact with sensitive data and critical functions, thereby maintaining the system's security and integrity.

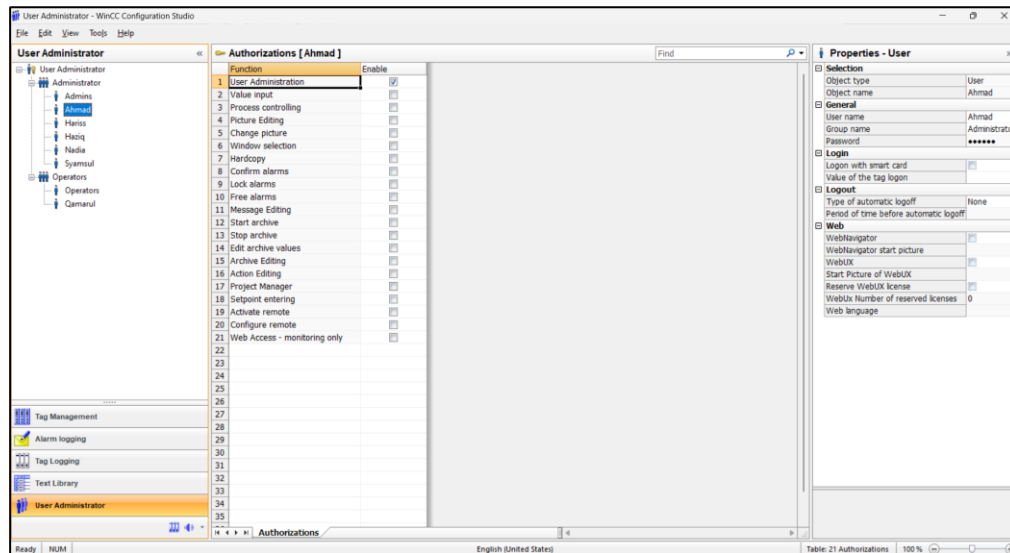


Figure 33 : Admin authorization



Figure 34 : System login in Log in layout

6.1.3 MONITORING AND CONTROL OF THE PROCESS

Here is a clear description of the SCADA system interface. It displays real-time monitoring of line production, including sensor readings and motor status. Additionally, it enables you to control each motor. By clicking on a motor, you can navigate to the motor screen and control it.

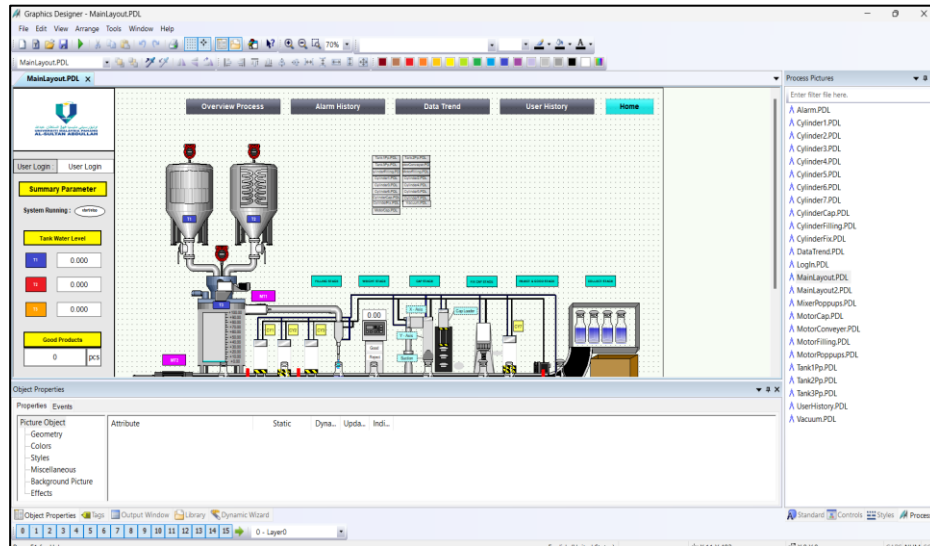


Figure 35 : Standards tools

The figure above are the tools to create a design of Automated Bottling Station visual.

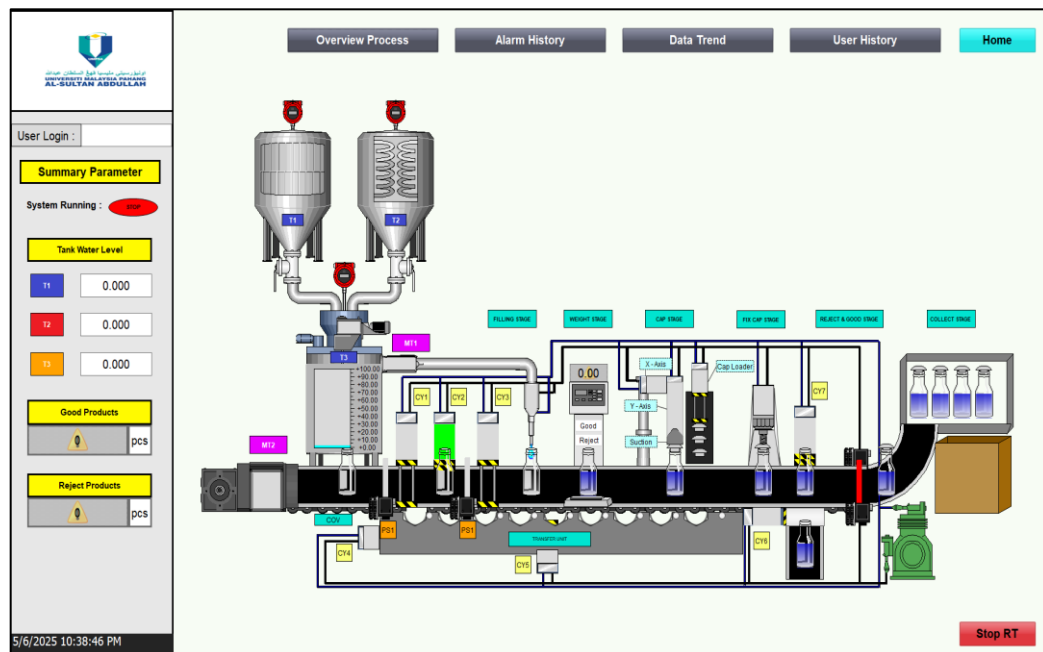


Figure 36 : Monitoring screen

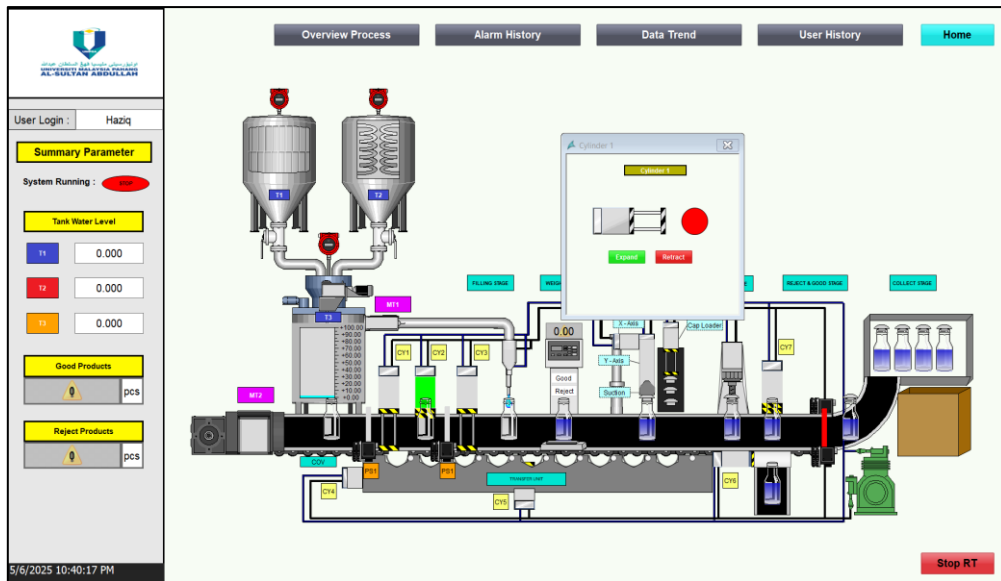


Figure 37 : Controlling screen

This is how to setup the sensor reading and motor controlling:

- i. **Create a tag address for each component that you want to use.**

Tag Management - WinCC Configuration Studio

File Edit View Tools Help

Tag Management

Internal tags

SIMATIC ST-1200, ST-1500 Channel

PLC

Structure tags

Actuator

Color

Motor

Tank

Weight

Tag Management

Alarm logging

Tag Logging

Text Library

User Administrator

Tags [OMS+]

Name	Data type	Length	Format adaptation	Connection	Group	Address	Linear scaling	AS value
1 ApprovedProductCount	Binary Tag	1		PLC2		M110.0		
2 Cylinder1Running(1200)Cov	Binary Tag	1		PLC		M40.6		
3 Cylinder2Running(1203)TULeft	Binary Tag	1		PLC		M41.2		
4 Cylinder3Running(1201)Cov	Binary Tag	1		PLC		M41.0		
5 Cylinder4Running(1204)TUFront	Binary Tag	1		PLC		M41.5		
6 Cylinder5Running(1202)Filling	Binary Tag	1		PLC		M41.7		
7 Cylinder6Running(1206)Cap	Binary Tag	1		PLC2		M101.6		
8 Cylinder7Running(1207)Cap	Binary Tag	1		PLC2		M101.4		
9 Cylinder8Running(1208)Fix	Binary Tag	1		PLC2		M102.2		
10 Cylinder8Running(1209)PushReject	Binary Tag	1		PLC2		M102.6		
11 Cylinder8Running(1313)GateReject	Binary Tag	1		PLC2		M102.5		
12 CylinderExpand1Start	Binary Tag	1		PLC		M40.7		
13 CylinderExpand2Start	Binary Tag	1		PLC		M41.1		
14 CylinderExpand3Start1203Left	Binary Tag	1		PLC		M41.3		
15 CylinderExpand4Start1204Front	Binary Tag	1		PLC		M41.4		
16 CylinderExpand5Start1202Fill	Binary Tag	1		PLC		M41.6		
17 CylinderExpand9PushReject	Binary Tag	1		PLC2		M103.0		
18 CylinderExpand10GateReject	Binary Tag	1		PLC2		M102.7		
19 DummyWaterFill	Unsigned 32-bit value	4	DwordToUnsignedDword	PLC				
20 EmergencyPB	Binary Tag	1		PLC2				
21 FlowmeterIndicator1	Binary Tag	1		PLC		M42.3		
22 FlowmeterIndicator2	Binary Tag	1		PLC		M42.4		
23 GoodProductCountW	Binary Tag	1		PLC		M1.5		
24 MixerMotor	Binary Tag	1		PLC		M42.5		
25 MotorFillingRunning	Binary Tag	1		PLC		M42.2		
26 OnMotorCap	Binary Tag	1		PLC2		M102.3		
27 PhotoDetect1	Binary Tag	1		PLC		M40.4		
28 PhotoDetect2	Binary Tag	1		PLC		M40.5		
29 RealValueWeight	Floating-point number 32-bit IEEE	4	FloatToFloat	PLC		M104		
30 RejectProductCount	Binary Tag	1		PLC2		M110.1		
31 RejectProductCountW	Binary Tag	1		PLC		M11.1		
32 RunMotorCap	Binary Tag	1		PLC2		M102.4		
33 RunningMotorCov	Binary Tag	1		PLC		M40.1		
34 runningvacuum	Binary Tag	1		PLC2		M101.4		
35 startCylinderExpand6_Cap1207	Binary Tag	1		PLC2		M100.7		

Ready NUM English (United States) Table: 46 Tags 100%

Figure 38 : Tag address

Set up the object by following the figures below. The remaining object is mostly like this setup.

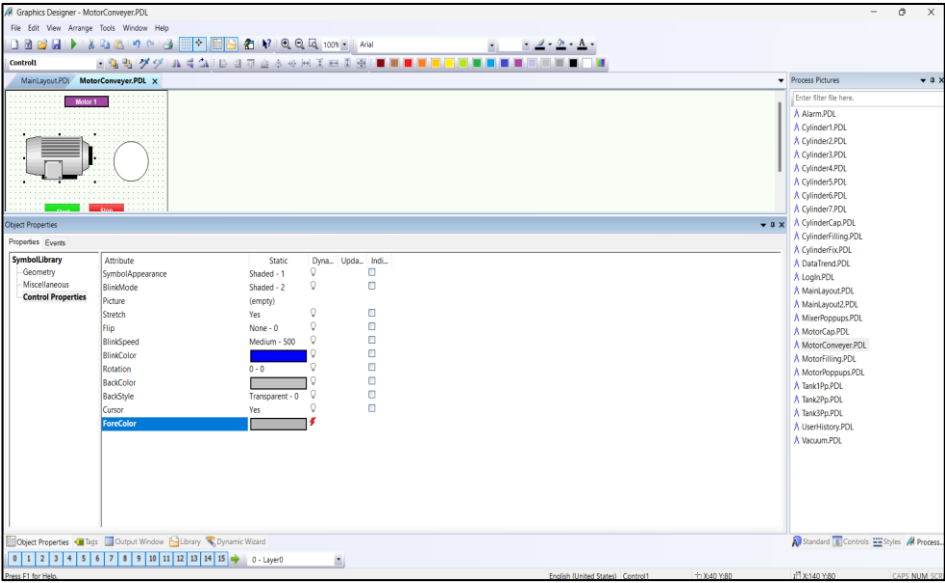


Figure 39 : Object properties 1

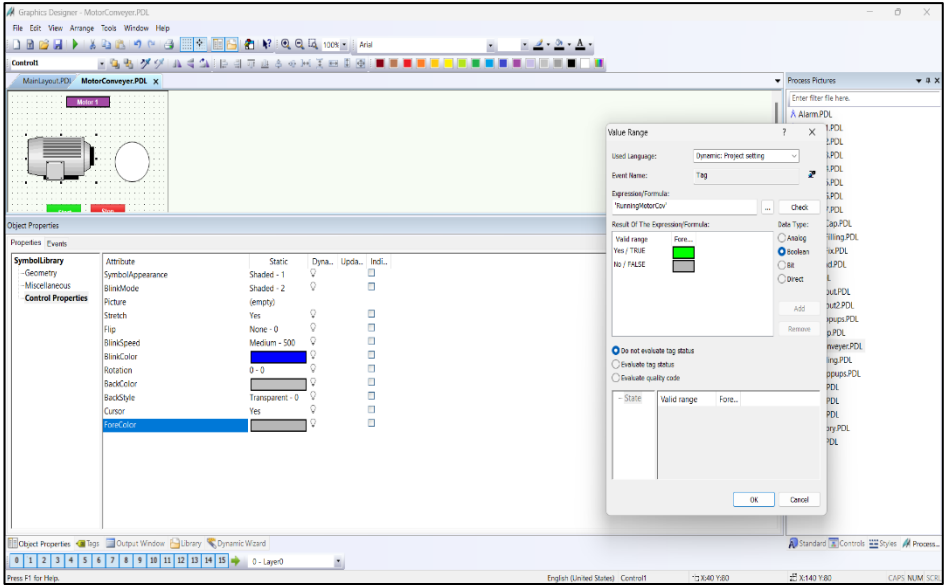


Figure 40 : Object properties 2

The figures below will show how to configure the button for controlling the motor.

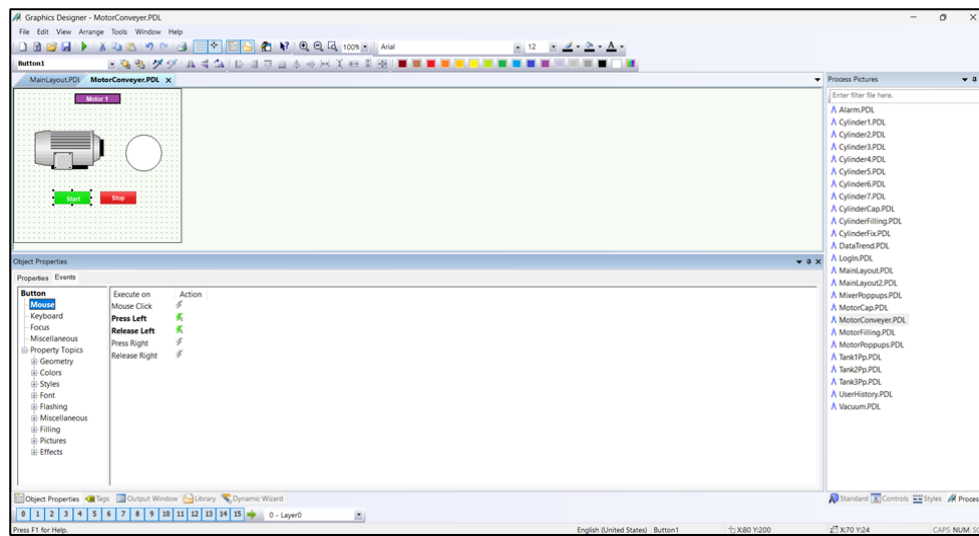


Figure 41 : Button configuration 1

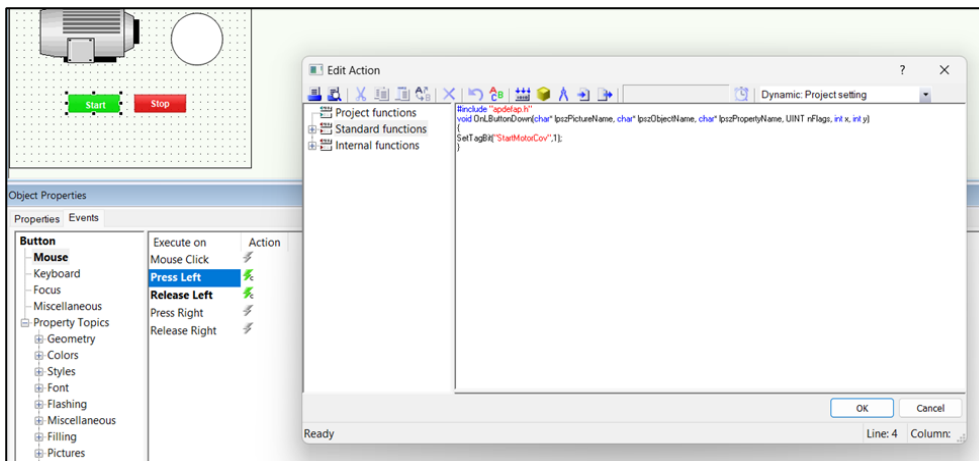


Figure 42 : Button configuration 2

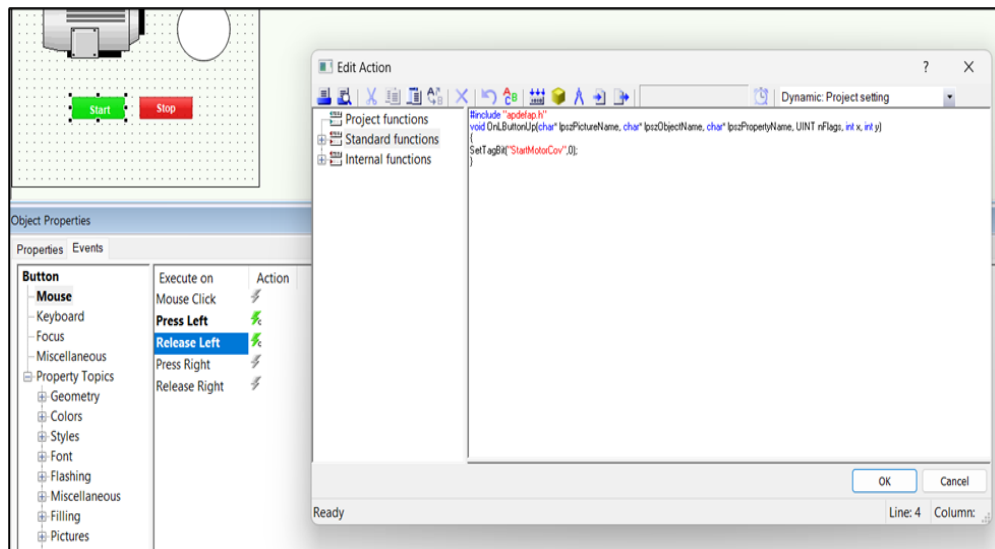


Figure 43 : Button configuration 3

6.1.4 ALARM HANDLING AND ACKNOWLEDGEMENT

Setting up a precautionary system is essential to detect any issues that may arise during the production process. The system should provide real-time monitoring and raise alarms to indicate if there is a problem. It should also distinguish between warnings, alarms, and failures and acknowledge them accordingly.

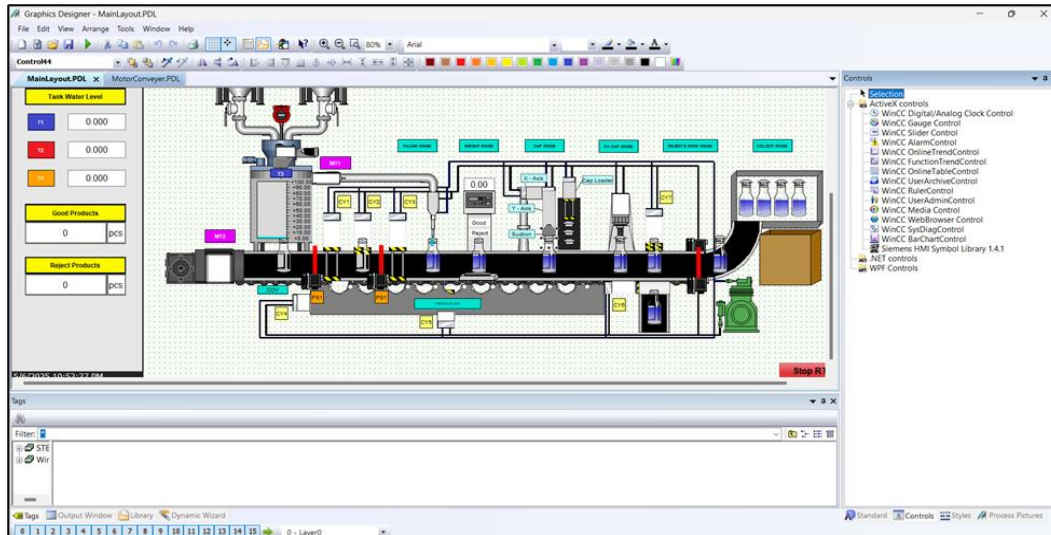


Figure 44 : Control for Alarm Control

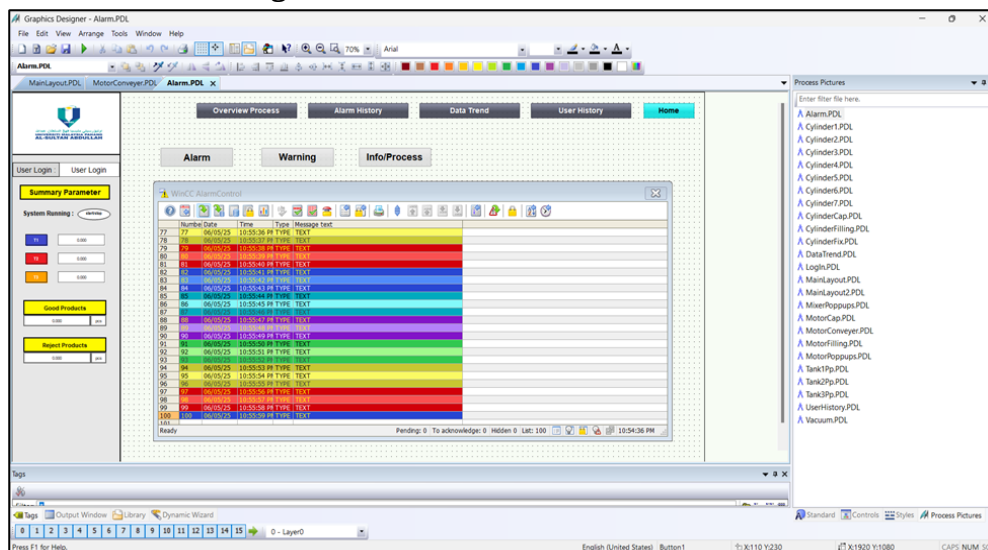


Figure 45 : Alarm Log Configuration

6.1.5 DATA VISUALIZATION AND REPORTING

On the SCADA main screen, all functional sensors such as flow, pressure, and temperature transmitters display corresponding data, such as tank level, water pressure, and temperature, respectively. The tank in the SCADA monitoring visual will show the level of water same as actual tank.

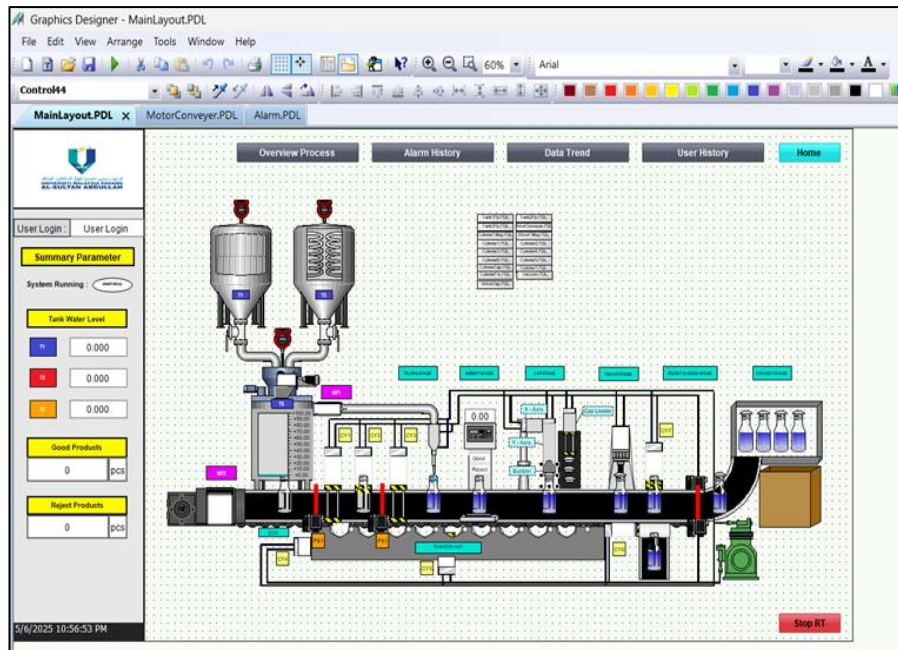


Figure 46 : Data visualization in monitoring screen

6.2 HMI OPERATION

6.2.1 INSTALLATION AND SETUP

i. System Requirements for TIA Portal Version 17

To ensure optimal performance, PC should meet the following system requirements:

- **Processor:** Intel Core i5 or AMD Ryzen 5000/7000 series with a frequency range of 2.5GHz to 4.2GHz.
- **RAM:** Recommended 16GB, with a minimum requirement of 8GB for basic functionality and 32GB for larger projects.
- **Hard Disk:** SSD with at least 15GB of free storage space for improved speed and performance.
- **Network:** Gigabit network speed for multi-user environments.
- **Operating System:** Windows 10 64-bit (Home edition) or compatible Windows Server versions.

ii. Installation Steps

1. Download the installation package from the Siemens website.
2. Mount the package and click on the "Start" application.
3. Choose the installation language (English is recommended).
4. Select the desired products to install (leave the default options).
5. Select the installation path and click "Next".
6. Accept the terms and conditions.
7. Click "Install" to start the installation process.
8. Wait for the installation to complete (estimated time: around 40-45 minutes).
9. Reboot your system when prompted.
10. After rebooting, insert the installation medium (DVD) if required.
11. Continue with the installation process and wait for it to complete.
12. Reboot your system again if prompted.

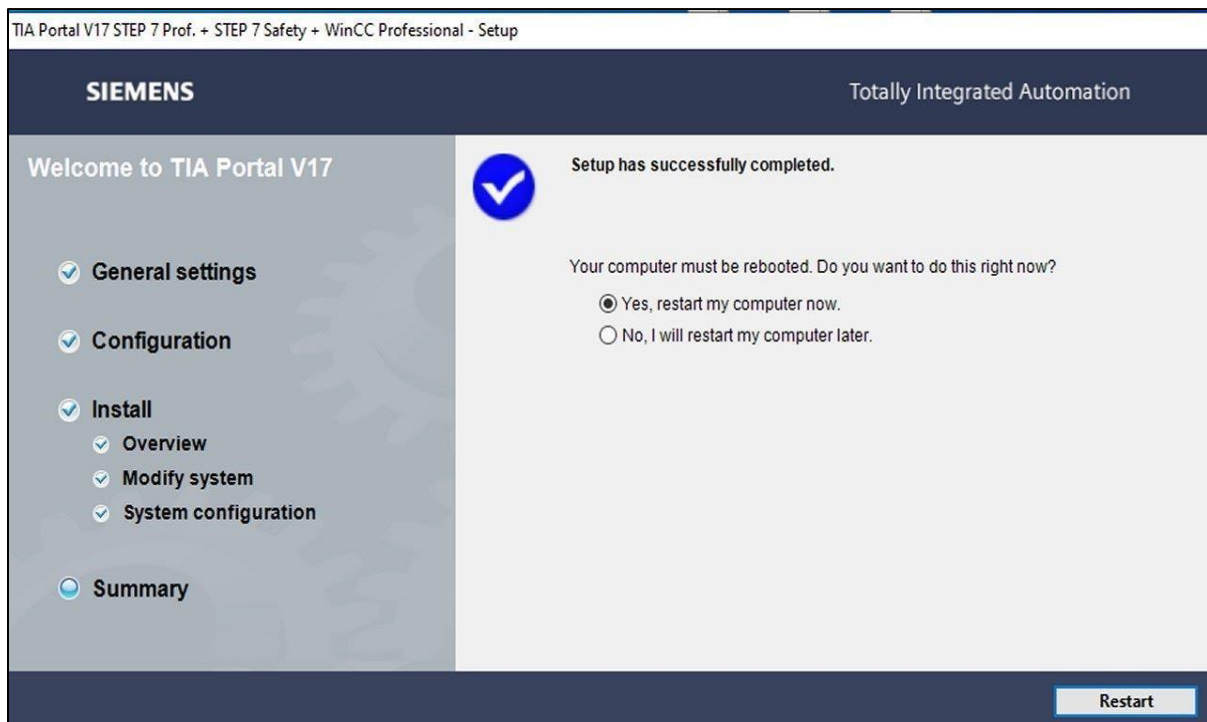


Figure 47 : Setup Complete on TIA Portal

6.2.2 LOGGING IN AND USER ACCESS CONTROL

Navigate to the **Users** tab to add or edit users like Admin, Operator, or Technician. You can set a password, choose how long before they are logged out, and put them into the right group like Administrator or Operator based on what they need to do.

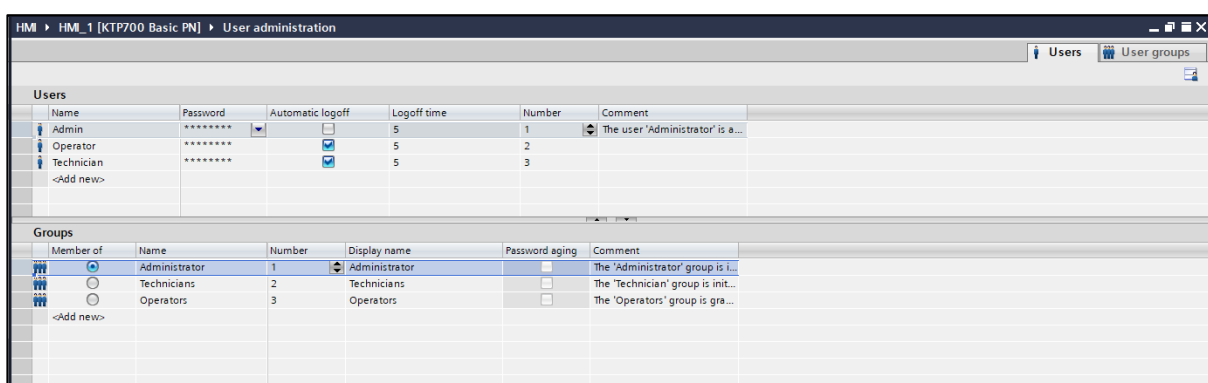
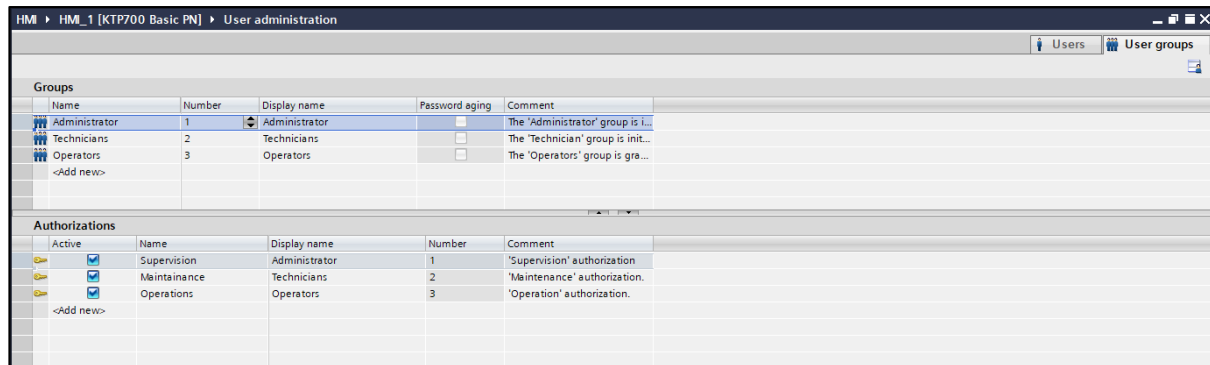


Figure 48 : User Administration

To manage user access to the system, go to User Administration. User groups with varying degrees of control, such as Administrator, Technician, and Operator, can be created here. Additionally, you can assign each group a login and password. This is helpful

to ensure that only authorised individuals may perform maintenance, make changes, or safely use the unit.



The screenshot shows the 'User administration' window in an HMI interface. It contains two main sections: 'Groups' and 'Authorizations'.

Groups Table:

Name	Number	Display name	Password aging	Comment
Administrator	1	Administrator	☐	The 'Administrator' group is i...
Technicians	2	Technicians	☐	The 'Technician' group is init...
Operators	3	Operators	☐	The 'Operators' group is gra...
<Add new>				

Authorizations Table:

Active	Name	Display name	Number	Comment
☒	Supervision	Administrator	1	'Supervision' authorization
☒	Maintenance	Technicians	2	'Maintenance' authorization.
☒	Operations	Operators	3	'Operation' authorization.
<Add new>				

Figure 49 : User Administration (User Group)

There are 3 groups of authorization which are supervision, maintenance and operations. Administrator group has the highest authority level that has all of the authorization levels which can monitor, control and manage the user administration. Technicians only have a maintenance and operations authority level which can monitor and control the system and for the guest, it's not have any authorization level in this system but it can login in the HMI on monitoring section only. Guests can only see the system running and cannot do any control to the system. This authority level is important in this system because we have to filter user access to improve the security of the system and this system cannot be accessed by random people.

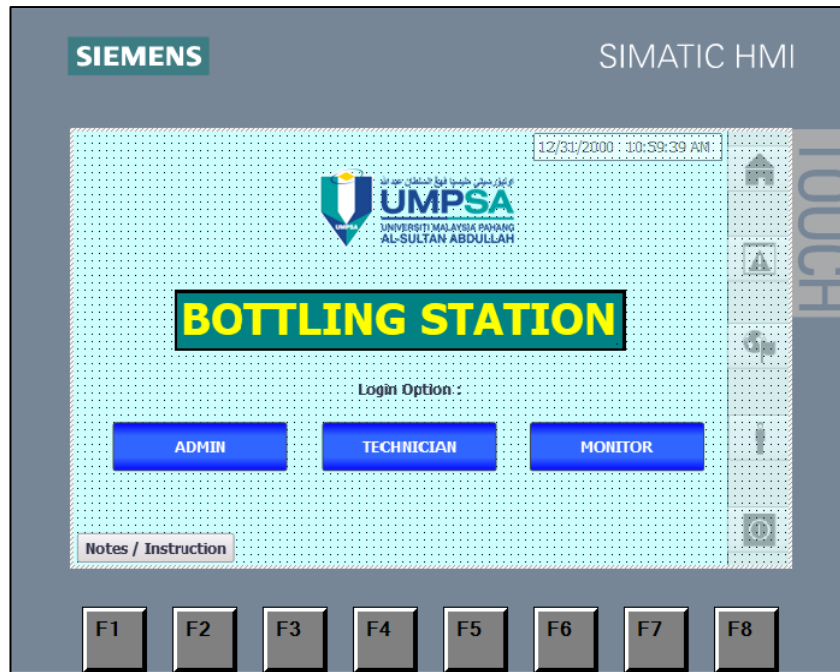


Figure 50 : Login Page (Start Screen)

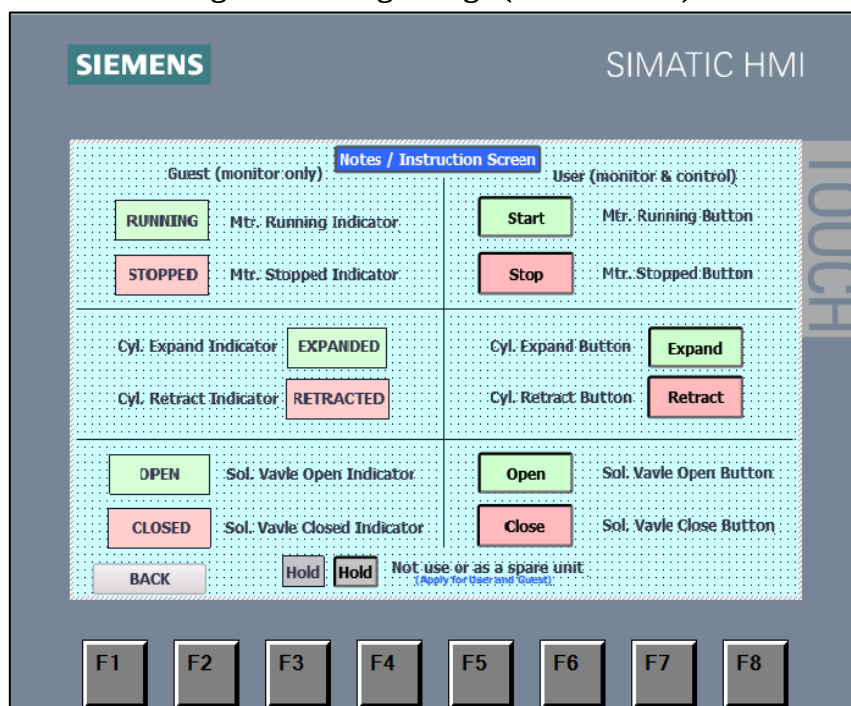


Figure 51 : Note / Instruction Page

This login page has a 3 login option that requires the user to login using their username and password for the Admin and Technician (assigned with different authorization levels). Guests can login using the monitor option without security and this screen has a note/instruction page for the user as a user guide in this HMI system.

6.2.3 DATA VISUALIZATION AND REPORTING

Home page acts as the main screen that can navigate users to each stage and have a general control of the system. In this Home page screen, the design of these 3 pages is the same but have a bit different of the user access can view in this page. Home page for admin and technician is almost the same but admin has a user administration navigation button and Guest only can open each stage to monitoring.

User Administration page shows all user information and Admin can make a change of each user in HMI Runtime which is their password, group and logoff time. For the **User page**, it can be accessed by Technician and Admin to control & monitor the system in Auto mode and Manual mode by each stage.

For the **Guest Stage**, it is accessible by anyone but for monitoring the system only and can't do any change or control. Some of the stages have a Hold parameter, this indicates not use or spare for future addition.

I. Home Page

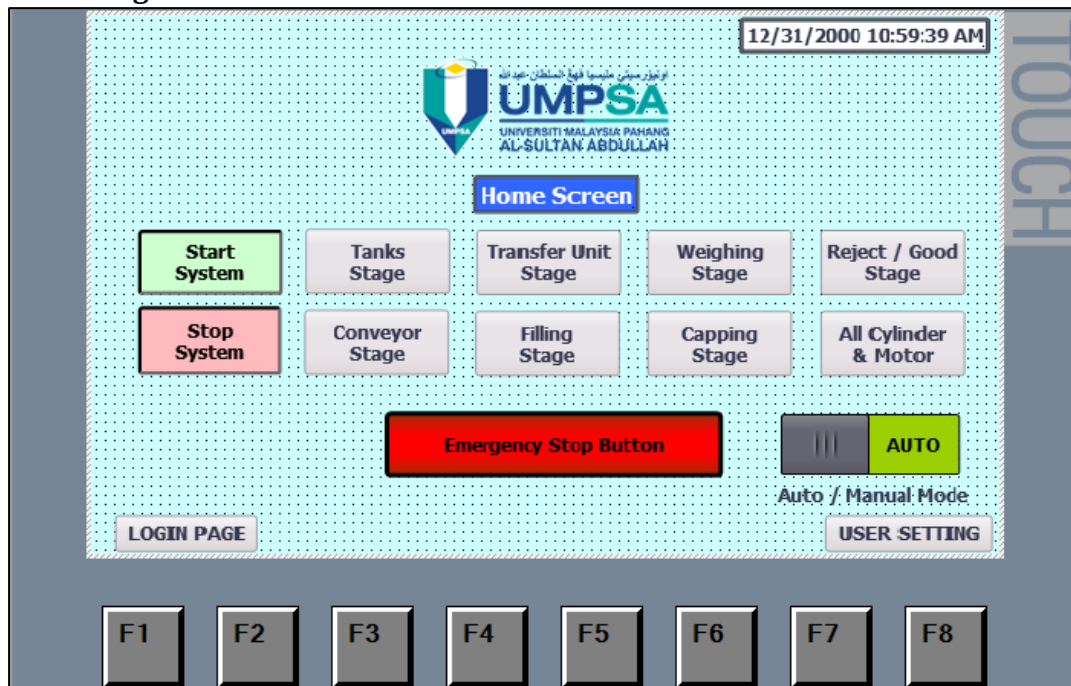


Figure 52 : Home Page(Admin)

This screen has a stage & user setting navigation button, auto/manual mode switch, general control (Start, Stop and Emergency Stop system in Auto mode)

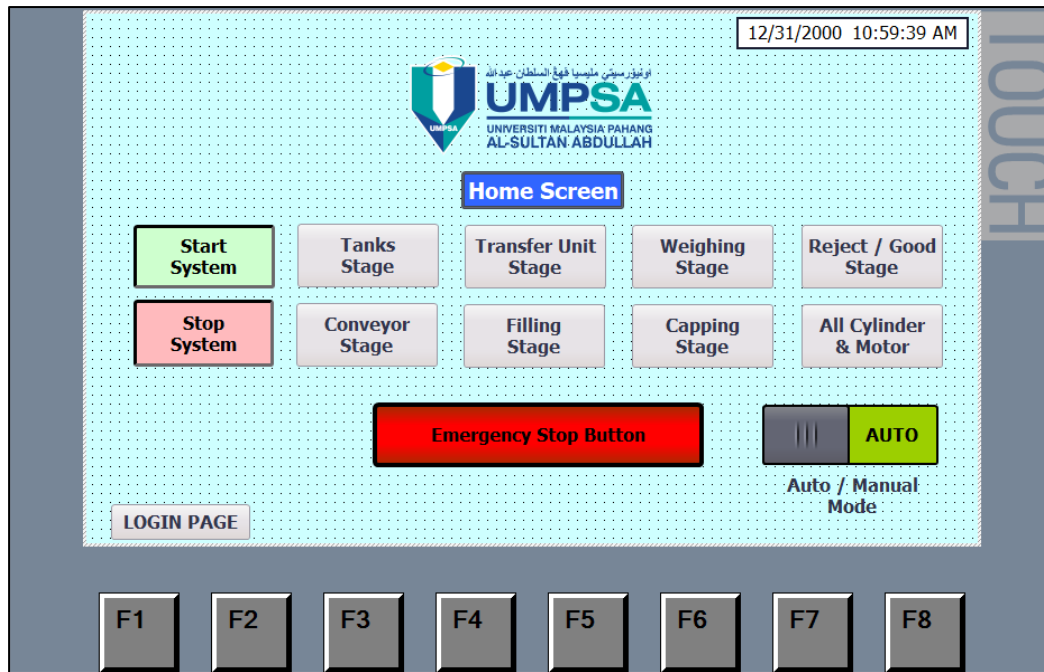


Figure 53 : Home Page(User)

This screen same with the previous but not allowed to open the user setting page because only and can change the user administration

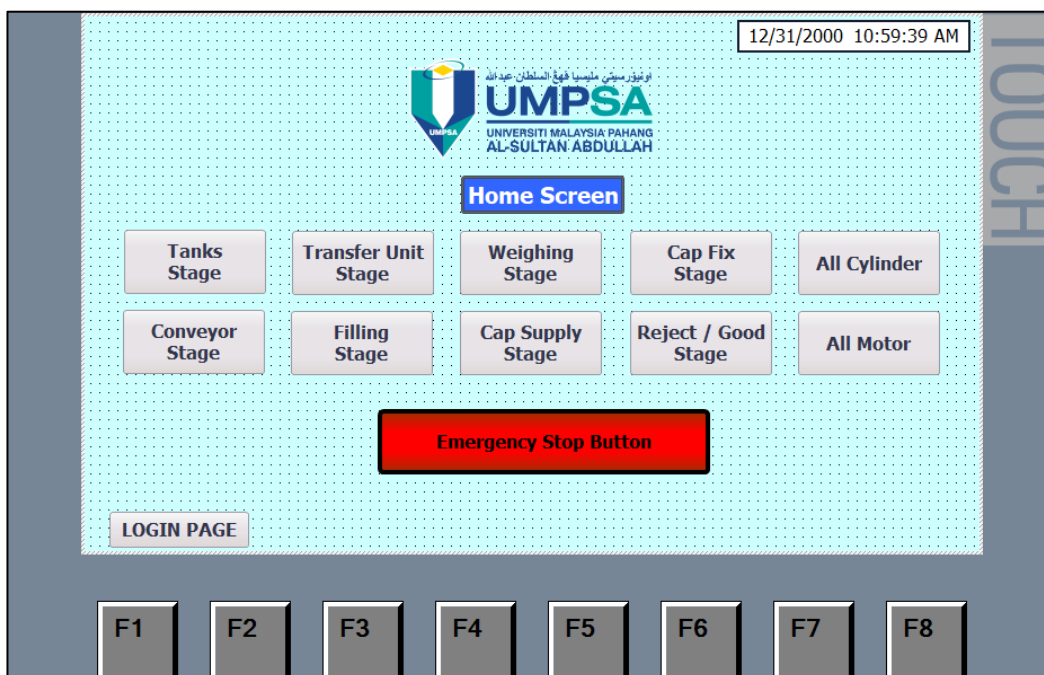


Figure 54 : Home page (Guest)

This screen only allowed user navigating to each stage just to monitoring and Emergency Stop the system only

II. User Administration Setting Page

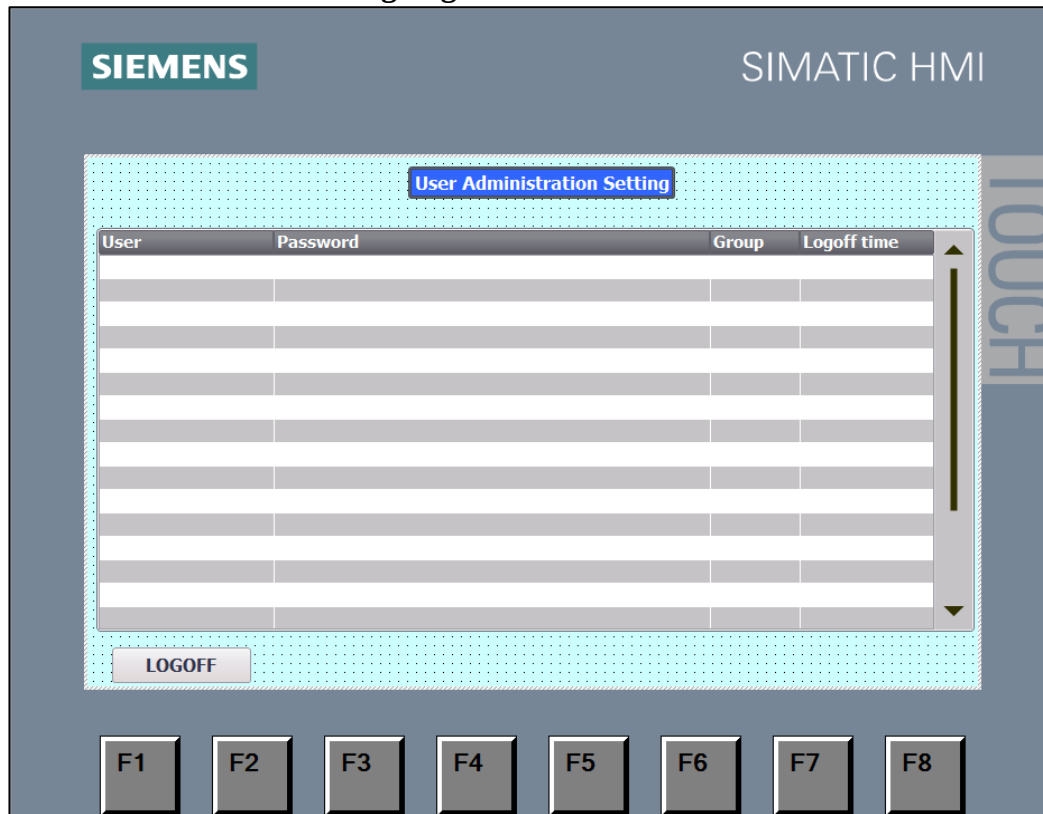


Figure 55 : User Administration Setting

This screen only can be opened by Admin to change the user password, their group of authority level and logoff time for all users and itself. Any changes on runtime are not saved to the Tia Portal but it's a good opportunity if you want to make quick changes without opening the software.

III. Tank Stage

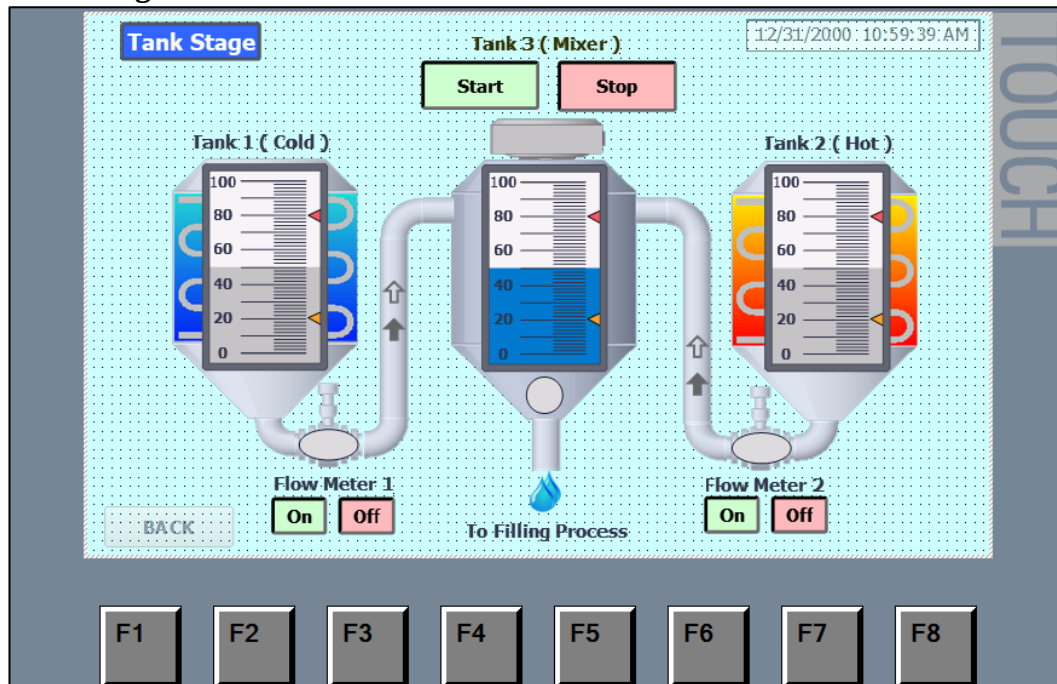


Figure 56 : Tank Stage (User)

This screen allow admin and technician to control the Mixer, Flowmeter 1 & 2 and monitor the current value of Mixer Tank Water Level.

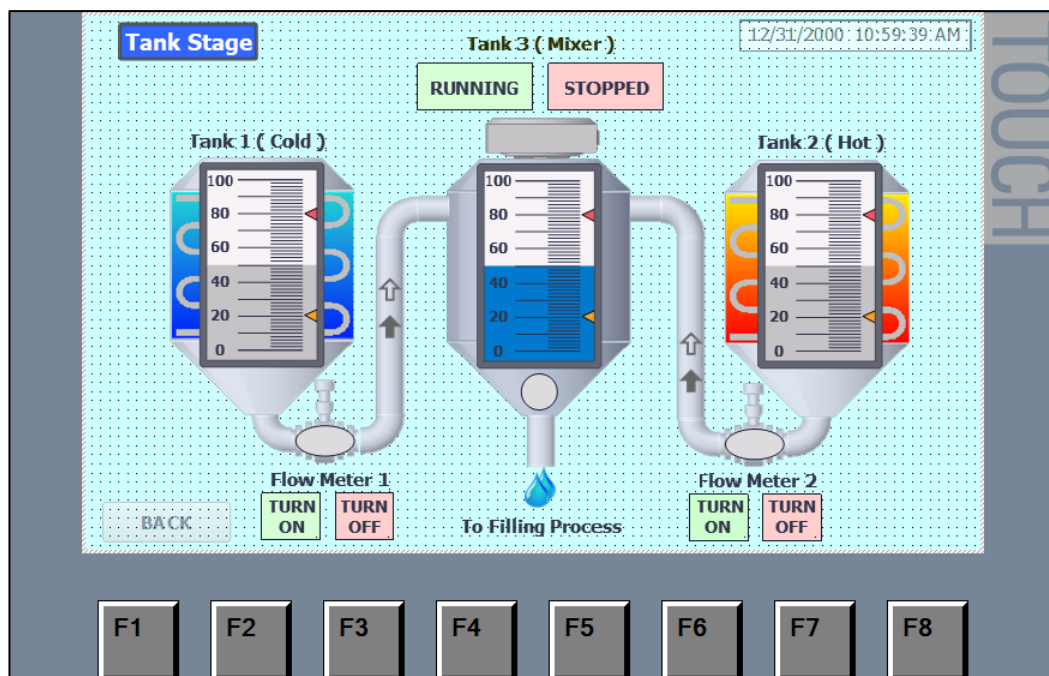


Figure 57 : Tank Stage (Guest)

This screen allow guest to monitor status of Mixer, Flowmeter 1 & 2 and current value of Mixer Tank Water Level

IV. Conveyor Stage

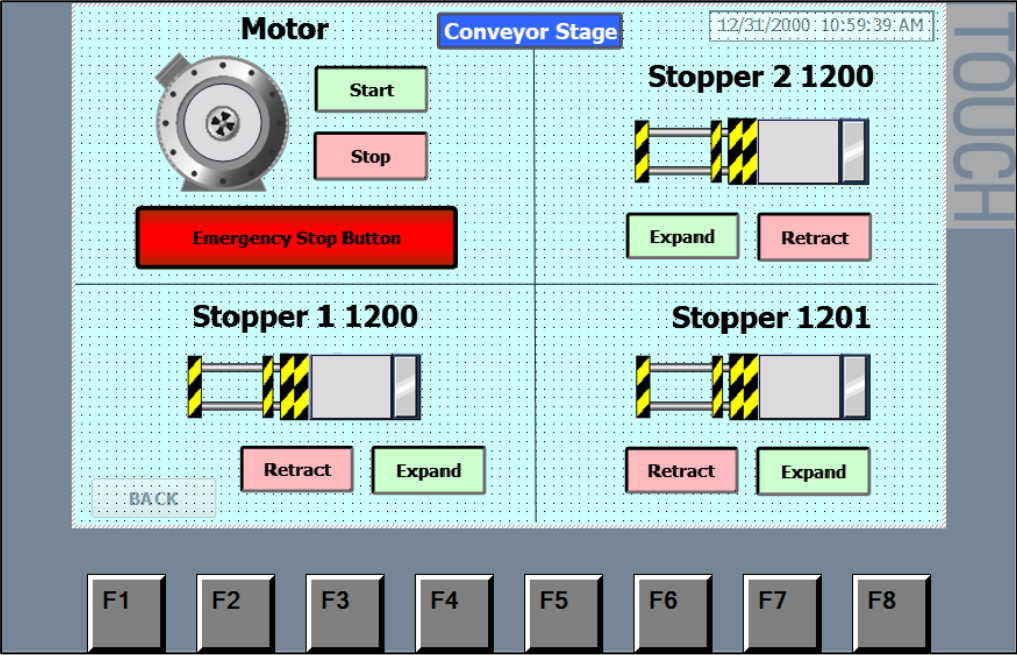


Figure 58 : Conveyor Stage (User)

This screen allow admin and technician to control Conveyor Motor, Stopper 1200 and Stopper 1201.

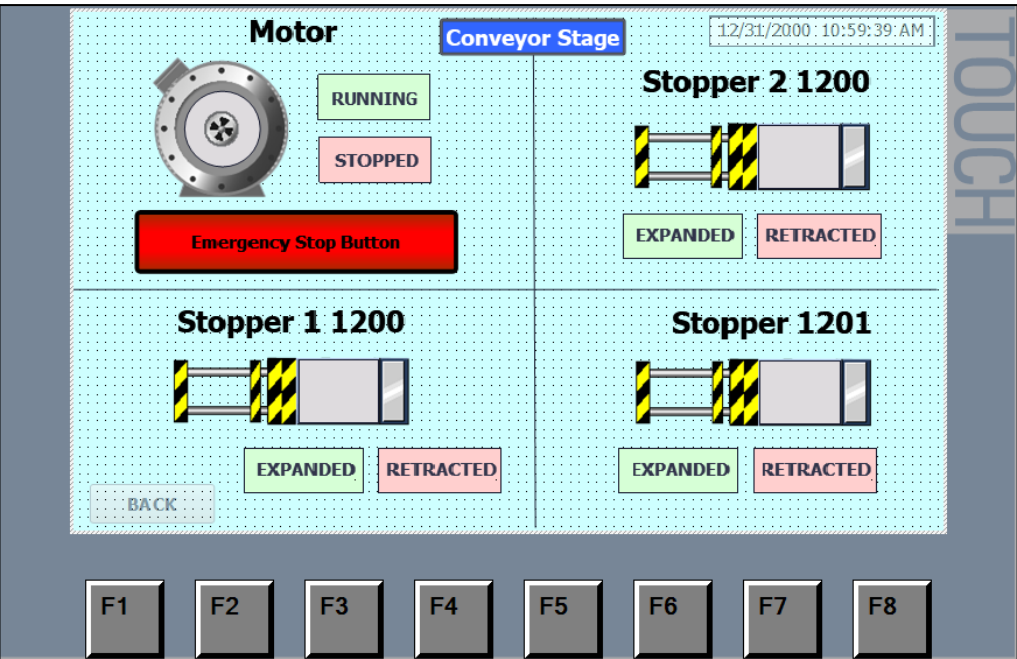


Figure 59 : Conveyor Stage (Guest)

This screen allow guest to monitor the status of Conveyor Motor, Stopper 1200 and Stopper 1201

V. Transfer Unit Stage

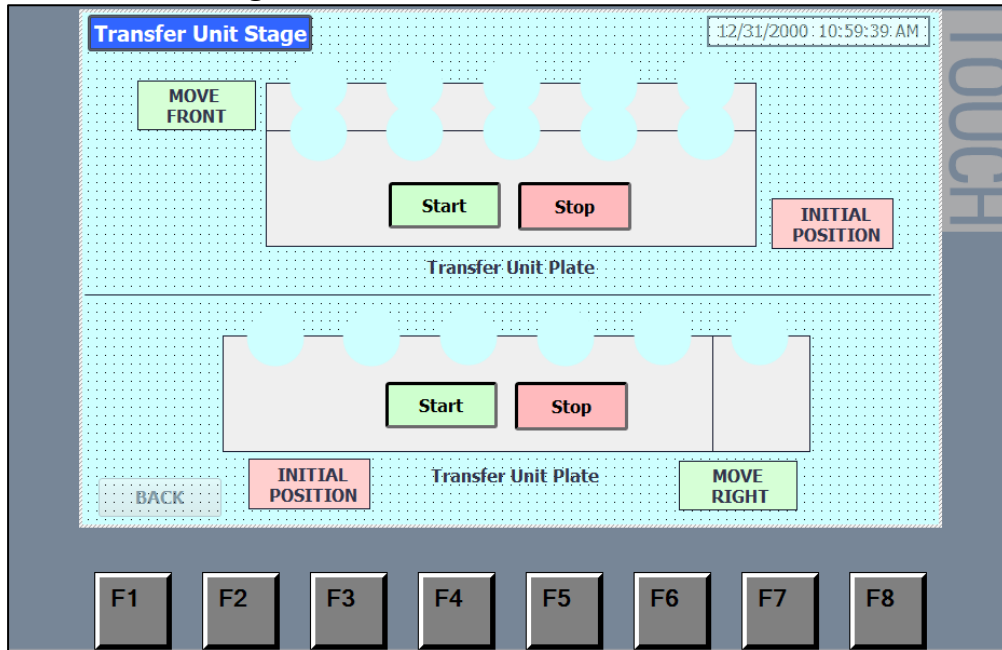


Figure 60 : Transfer Unit (User)

This screen allow admin and technician to control cylinder on Transfer Unit Stage which is 1206 (left) and 1207 (front)

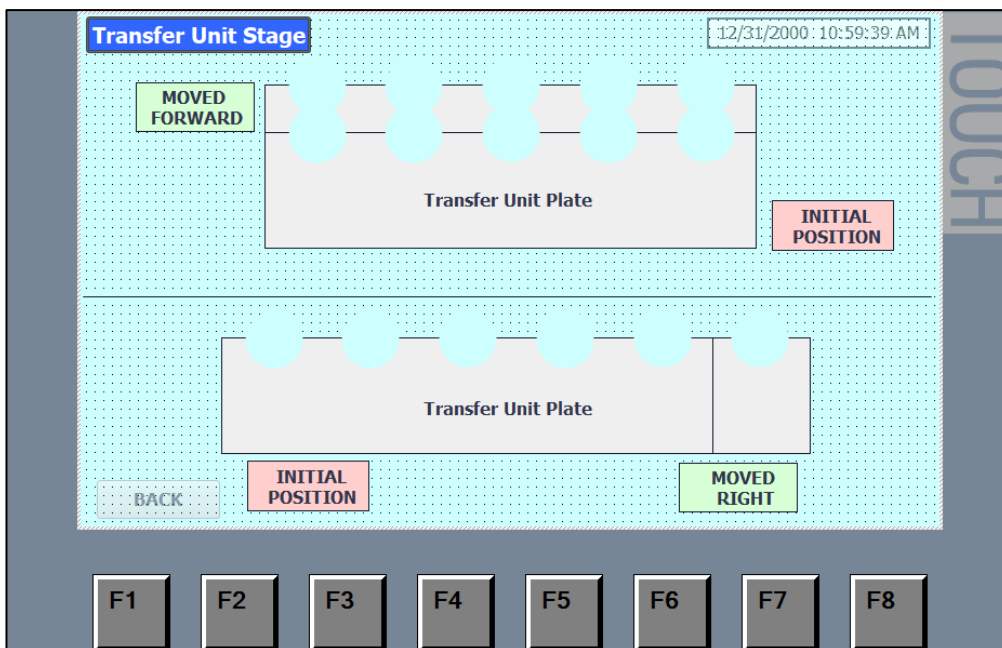


Figure 61 : Transfer Unit (Guest)

This screen allow guest to monitor the status of cylinder on Transfer Unit Stage which is 1206 (left) and 1207 (front).

VI. Filling Stage

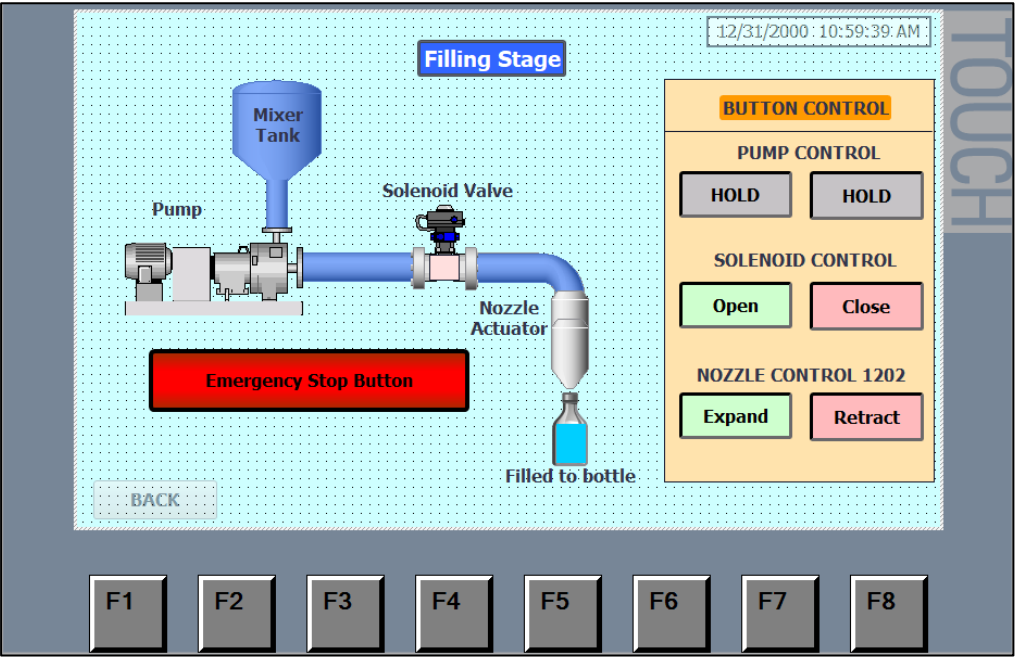


Figure 62 : Filling Stage (User)

This screen allow admin and technician to control the motor pump, solenoid valve and filling nozzle

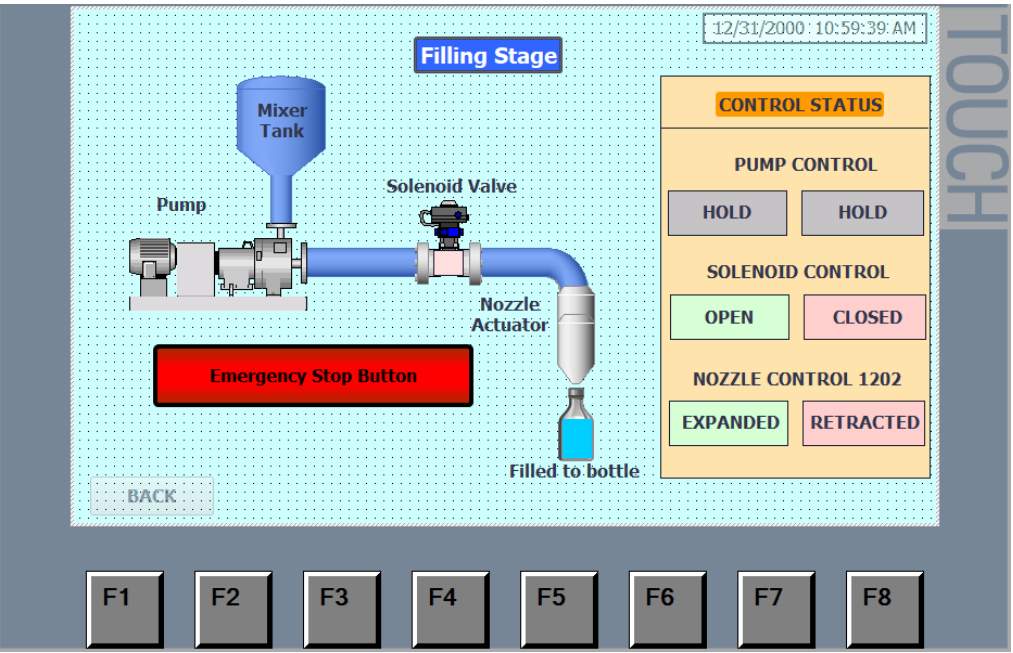


Figure 63 : Filling Stage (Guest)

This screen allow guest to monitor the status of motor pump, solenoid valve and filling nozzle

VII. Weighing Stage

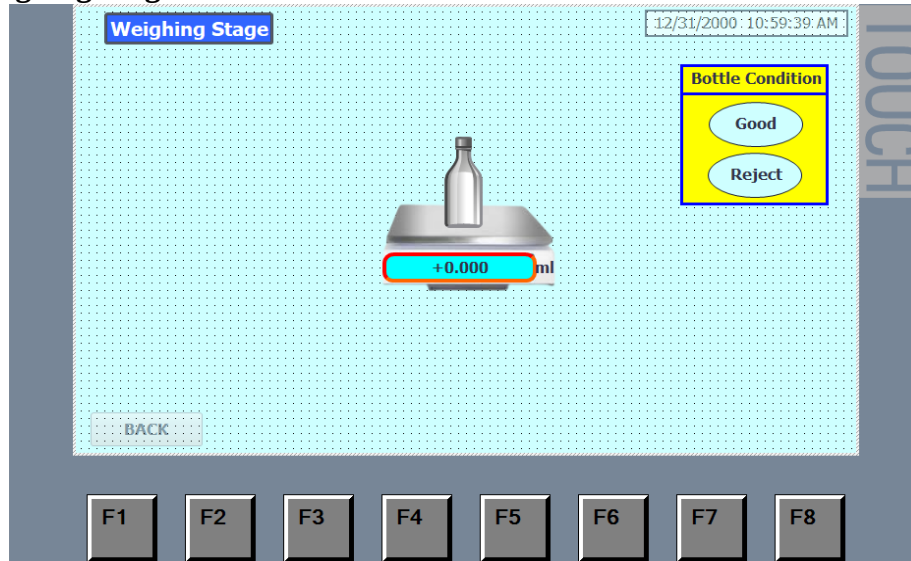


Figure 64 : Weighing Stage (User & Guest)

This screen allow admin, technician and guest to monitor the current value of the bottle weight and the bottle condition are good or rejected.

VIII. Cap Supply Stage

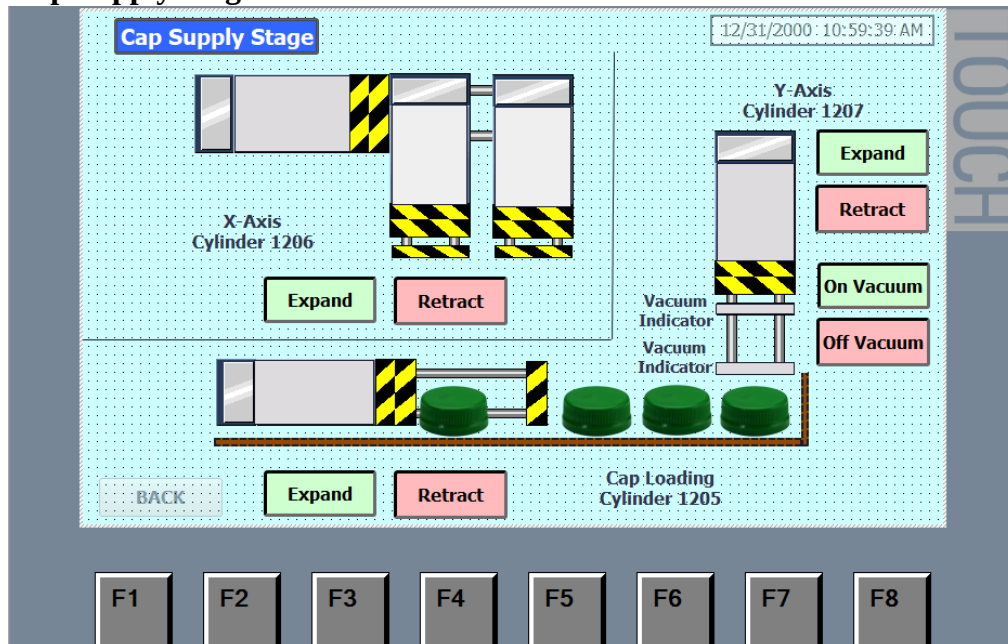


Figure 65 : Cap Supply Stage (User)

This screen allow admin and technician to control cylinder x-axis 1206, cylinder y axis 1207, cap loading cylinder 1205 and vacuum suction

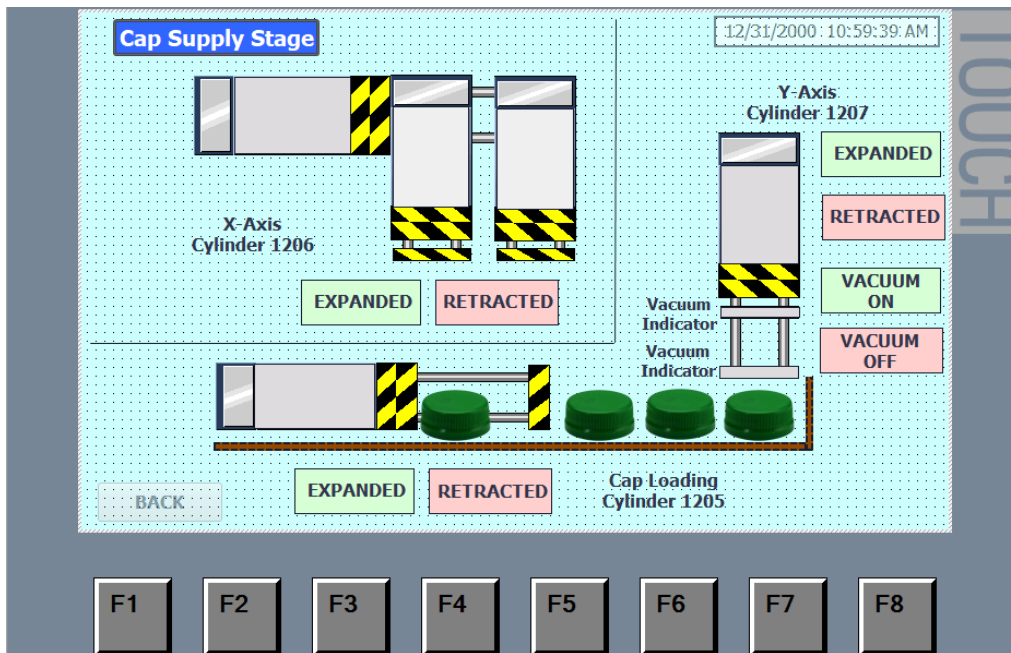


Figure 66 : Cap Supply Stage (Guest)

This screen allow guest to monitor the status of x axis cylinder 1206, y axis cylinder 1207, cap loading cylinder 1205 and vacuum suction.

IX. Cap Fix Stage

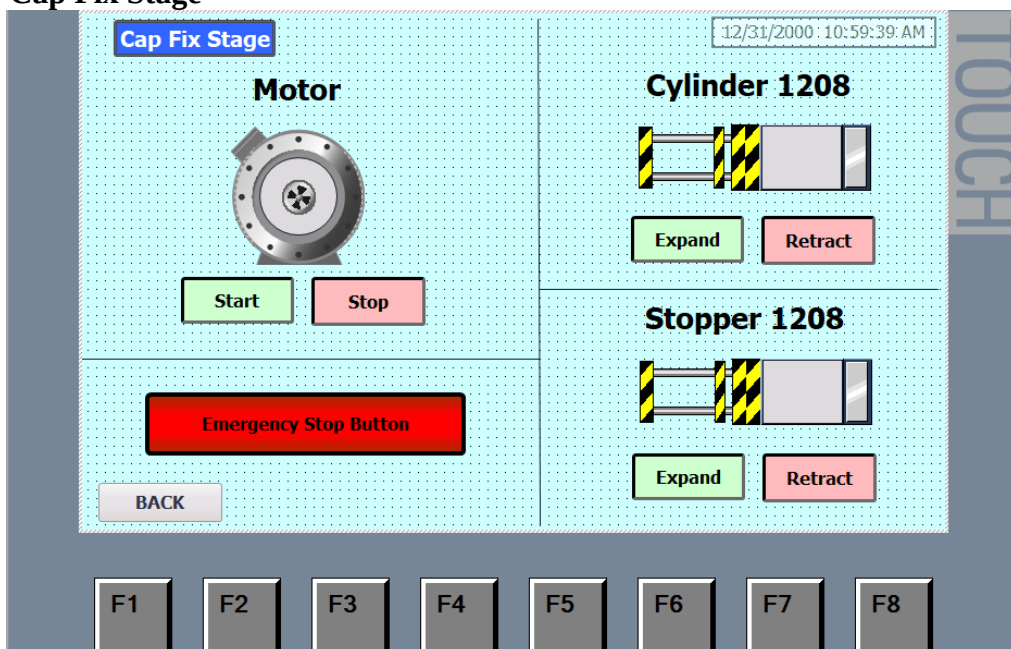


Figure 67 : Cap Fix Stage (User)

This screen allow admin and technician to control cap fix motor, motor position cylinder and bottle stopper 1208.

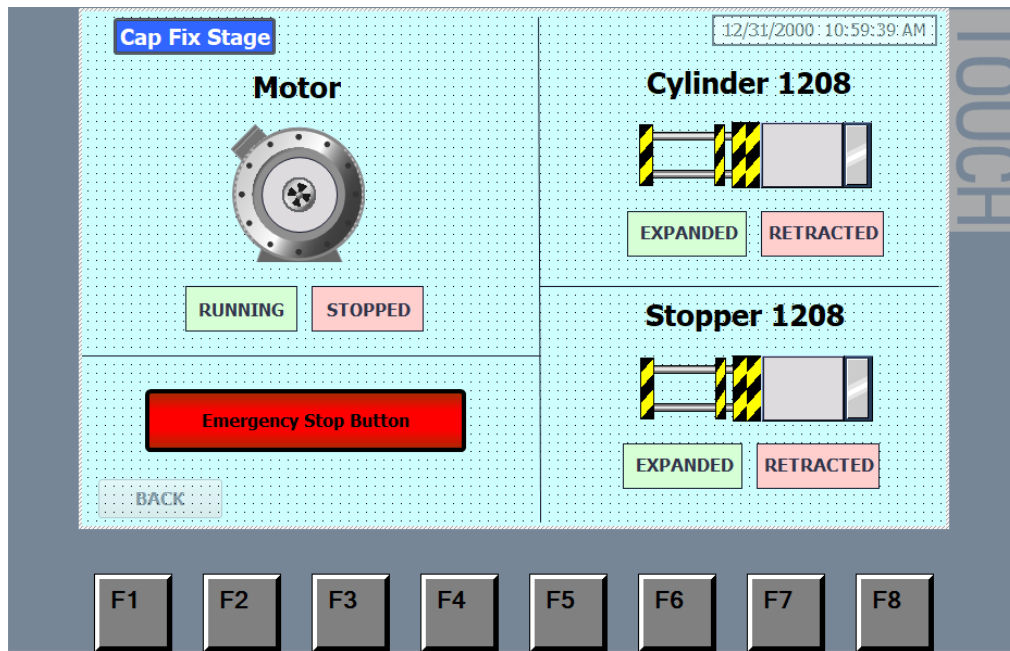


Figure 68 : Cap Fix Stage (Guest)

This screen allow guest to monitor status of cap fix motor, cylinder motor position and bottle stopper 1208.

X. Reject / Good Stage

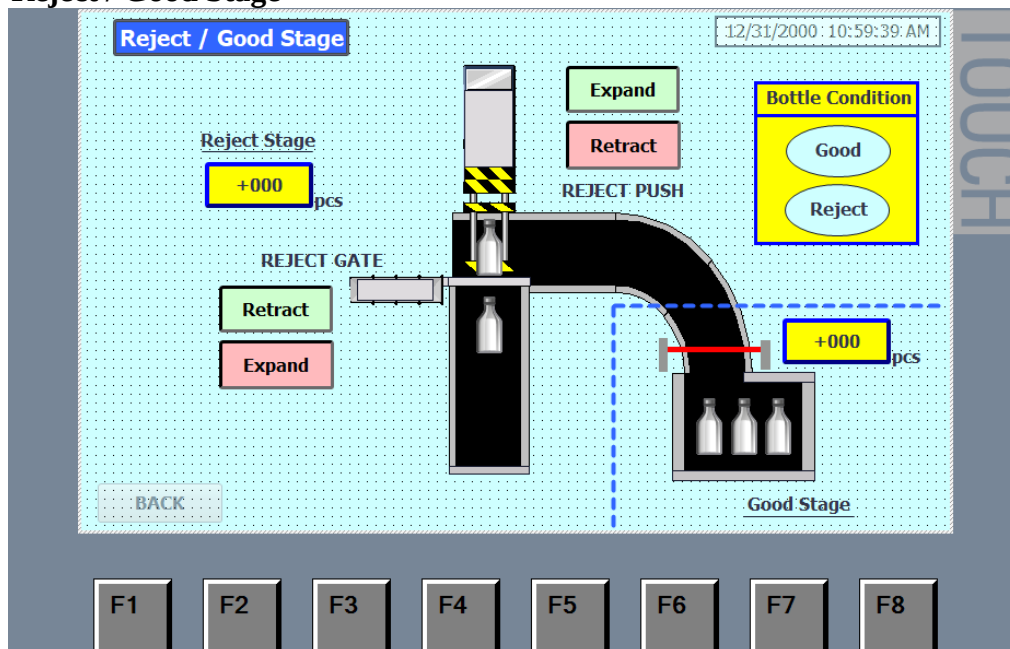


Figure 69 : Reject / Good Stage (User)

This screen allow admin and technician to control reject gate and reject pusher. Also monitoring the status and current value of good and rejected products.

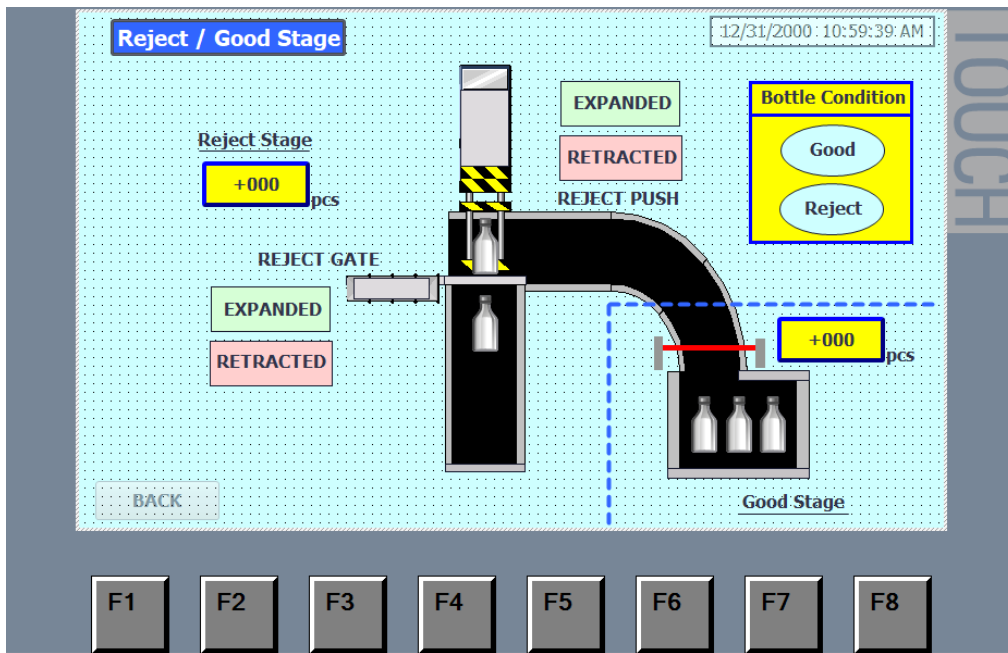


Figure 70 : Reject / Good Stage (Guest)

This screen allow guest to monitor the status of reject gate, reject pusher, good and reject bottle also current value of good and rejected product.

XI. AllCylinder1

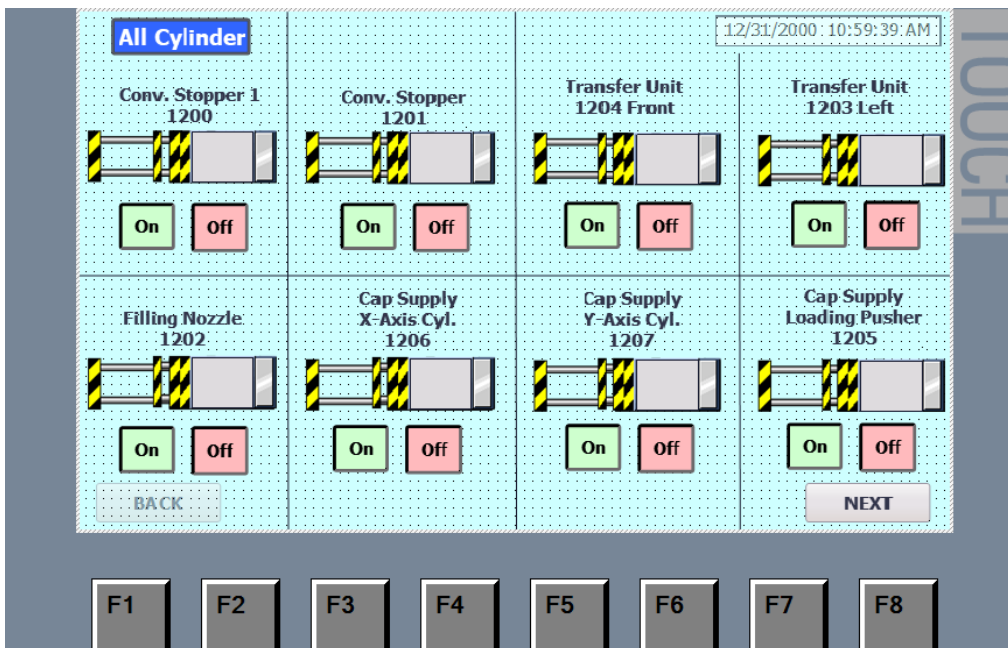


Figure 71 : All Cylinder 1 (User)

This screen allow admin and technician to control all cylinder in the system to ease the troubleshooting and manual testing off all cylinder from conveyor stage into cap supply stage.

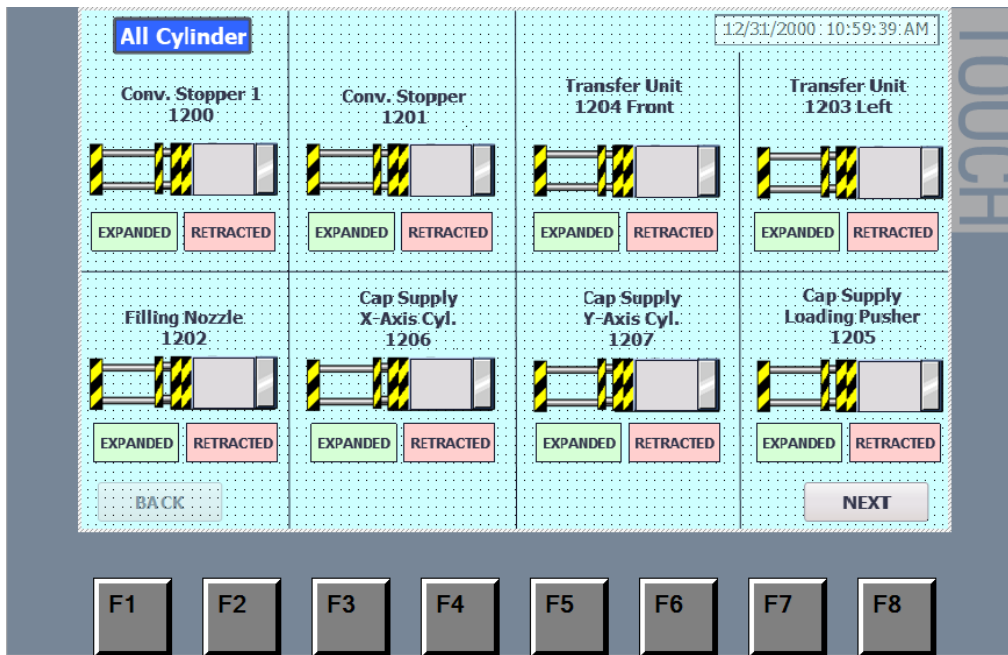


Figure 72 : All Cylinder 1 (Guest)

This screen allow guest to monitor all cylinder in the system to ease the troubleshooting and manual testing off all cylinder from conveyor stage into cap supply stage.

XII. AllCylinder2

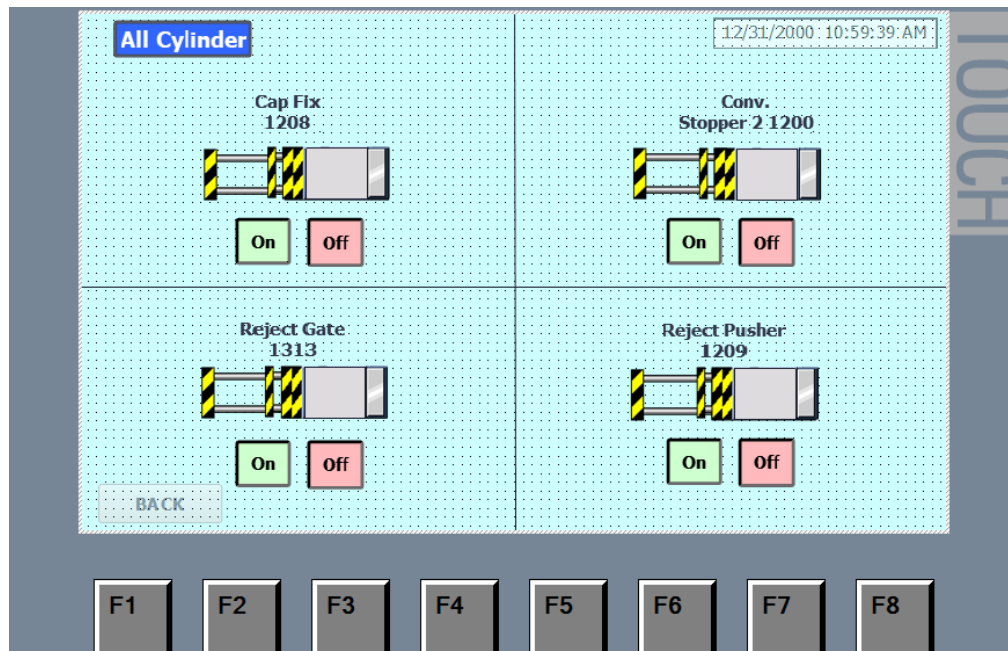


Figure 73 : All Cylinder 2 (User)

This screen allow admin and technician to control all cylinder in the system to ease the troubleshooting and manual testing off all cylinder from cap fix stage stage into reject stage stage.

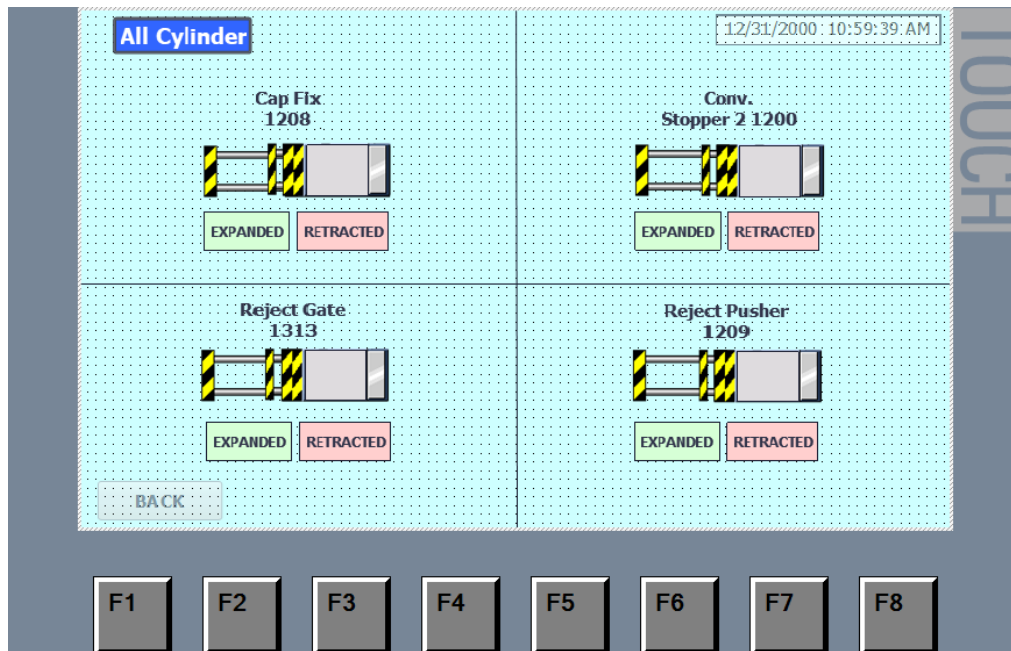


Figure 74 : All Cylinder 2 (Guest)

This screen allow guest to monitor all cylinder in the system to ease the troubleshooting and manual testing off all cylinder from cap fix stage stage into reject stage stage.

XIII. All Motor

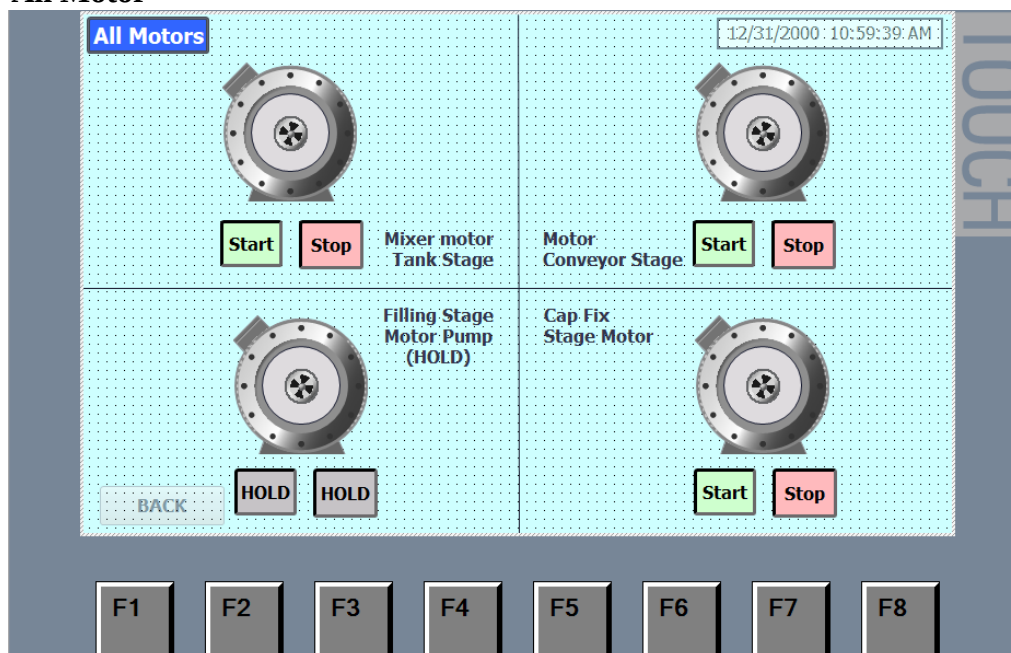


Figure 75 : All Motor (User)

This screen allow admin and technician to control all motor in the system to ease the troubleshooting and manual testing off all motor in one page.

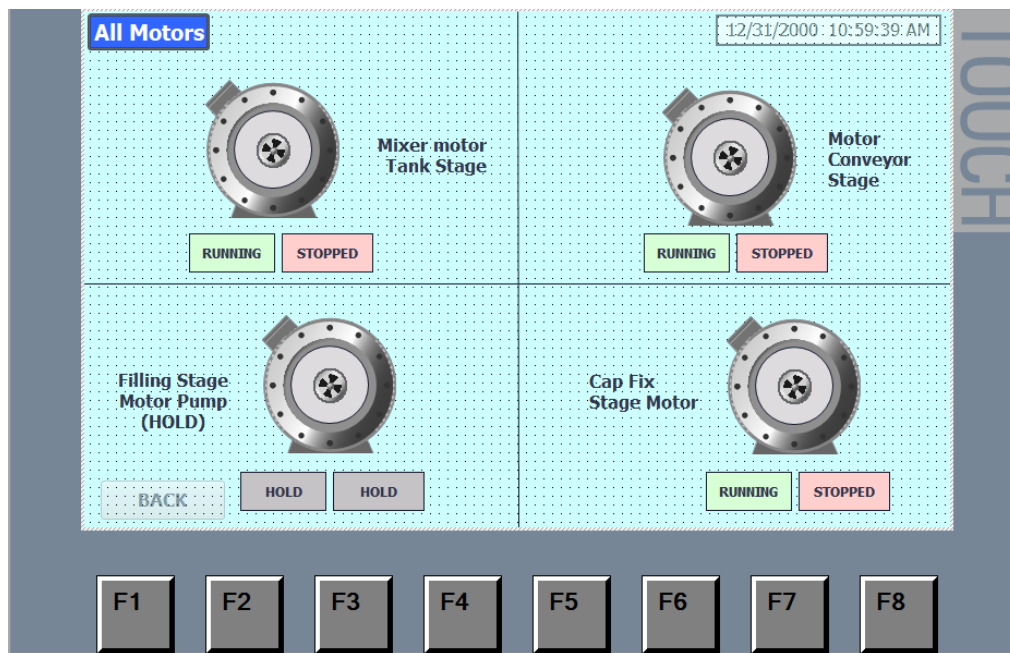


Figure 76 : All Motor (Guest)

This screen allow guest to monitor all motor in the system to ease the troubleshooting and manual testing off all motor in one page.

XIV. Page Template

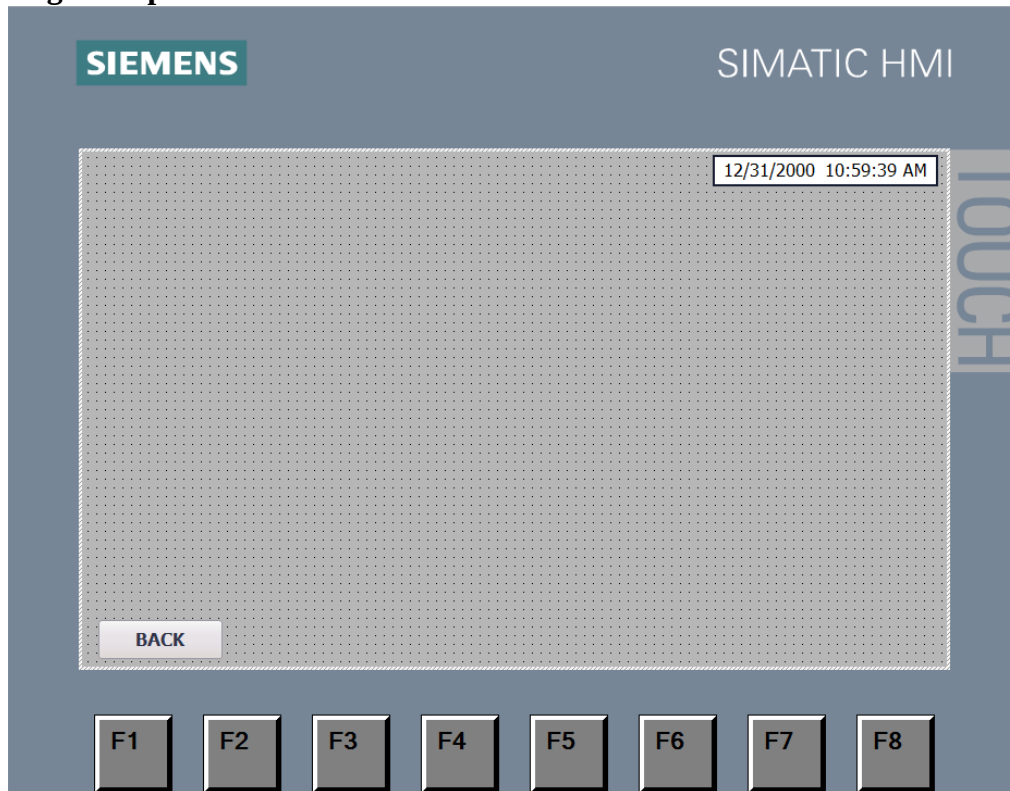


Figure 77 : Template (User & Guest)

This screen is used for a screen template of all user and guest screen to ease the hmi development without copying the back button and date time to all screen

7.0 MAINTENANCE AND TROUBLESHOOTING

7.1 SCADA

7.1.1 SCADA SYSTEM INTEGRATION ISSUE

During the SCADA development process for the bottling station, we faced a critical issue where the SCADA system (WinCC Explorer) could not communicate with the Siemens S7-1200 PLC. Upon investigation, the root cause was found to be an incorrect TCP/IP configuration between the PLC and the SCADA. Specifically, the IP address and subnet mask of the PLC were not properly aligned with the SCADA communication settings.

To resolve the issue, we verified and matched the IP address of the PLC in the TIA Portal hardware configuration with the communication settings in WinCC. We also ensured that both devices were connected on the same physical network and had matching subnet configurations. After correcting the TCP/IP settings and performing a successful PING test from the SCADA system to the PLC, communication was successfully established.

7.2 HMI

The Human Machine Interfaces (HMI) confronted two issues involving hmi tag not written to the plc and reject & good counted value are not displayed on the i/o field. Hmi tag not written to the plc is caused because we are using a physical address to integrate with the hmi. This is because hmi can't read or write the physical address and hmi has no authority to override them. The solution of this issue is we use a memory address to control the logic and we make sure the hmi tag connections are connected to the right plc tag and also ensure all the hmi logic are connected parallelly with the physical output.

Next, the counted value of reject and good product are not displayed on the i/o field because we use the wrong type of memory address which is MD. This is because the MD with a Real data type is storing floating-point value, so it causes data type mismatch. The solution is, we use a MW with Word data type instead of using MD with Int data type because word will always count a positive value and int may count the negative value that will show blank. The i/o field works after we fix it but still have a small issue which is the displayed value is not holding but it flickering continuously, it displays current value then changes to zero.

All the issues stated above are solved and HMI integration works well in this Water Bottling Station, also it can be used for the Auto and Manual mode separately without bothering each other plc sequence.

8.0 DISCUSSION & FACED CHALLENGES

8.1 Analog Water Level Flickering Issue

While developing the SCADA system for the bottling station, we encountered a challenge where the water level reading (stored in memory address %MD105) was unstable, constantly flickering between 0 and the actual value on both the SCADA and HMI screens. Although the sensor was functioning correctly, the issue stemmed from the PLC and SCADA accessing the same memory address simultaneously, leading to data clashes. This was further exacerbated by electrical noise from the analog signal (4–20mA converted to 0–10V) and rapid SCADA polling intervals of 100ms.

To resolve this, we proposed using a MOV instruction in the PLC to copy the value from %MD105 to a new memory address (%MD110), allowing SCADA and HMI to read from %MD110 instead, thus avoiding direct memory access conflicts. Additionally, we recommended implementing analog filtering techniques such as moving average or low-pass filters to mitigate signal noise, verifying proper wiring grounding and shielding, adjusting SCADA polling intervals to 250–500ms for smoother updates, and formatting the SCADA display to show only 1–2 decimal places to reduce the appearance of flickering.

8.2 CTU Counter Display Flickering Issue

Another issue we faced involved the bottle counters on the SCADA screen—Approved, Rejected, and Total Bottles—where the values either updated slowly, flickered between 0 and the actual count, or occasionally failed to update altogether. This problem primarily arose due to the use of incorrect data types in SCADA, where 16-bit tags were used for counters that required 32-bit, and some memory addresses overlapped (e.g., %MW30 and %MD31), causing data conflicts when multiple counters accessed the same memory area. To address this, we recommended using appropriate data types: INT (16-bit) for smaller values like rejected bottles (%MW30) and DINT (32-bit) for larger values like approved (%MD31) and total bottles (%MD33).

We also suggested assigning separate memory addresses for each counter and using a MOV instruction to transfer the actual counter values to buffer memory before passing them to SCADA, thereby avoiding conflicts. Additional measures included adjusting SCADA polling intervals to 250–500ms and formatting the display to show whole numbers only, ensuring a more stable and readable output.

9.0 SAFETY GUIDELINE

9.1 SCADA WinCC 7.5

Safety Notes

This manual contains information that must be observed to ensure your personal safety and to prevent property damage. Notices referring to your personal safety are highlighted in the manual by a safety alert symbol; notices referring to property damage only have no safety alert symbol. Depending on the hazard level, warnings are displayed in a descending order as follows:

 DANGER
means that there can be severe physical injury or even death if the corresponding safety measures are not followed.
 WARNING
means that there can be severe physical injury or even death if the corresponding safety measures are not followed.
 CAUTION
indicates that minor personal injury may result if proper precautions are not taken.
NOTICE
indicates that damage to property may result if proper precautions are not taken.

Note:

It is an important information about the product, the way to handle the product or the respective part of the documentation and we wish to especially bring this to your notice.

If multiple levels of hazards can occur, the warning is always displayed with the highest possible level. If a warning with a warning triangle is to be indicate physical injury, the same warning may also contain information about damage to property.

Qualified Personnel

The corresponding machine/ system may only be set up and operated with the help of this documentation. A device/system must only be commissioned and operated by qualified personnel. Qualified persons in the sense of safety instructions in this documentation stand for persons who are authorized to operate, earth and mark machines, system and electrical circuits according to safety standards.

Proper use

Please observe the following:

Brands

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Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, solutions, machines, equipment and/or networks. They are important components in a holistic industrial security concept. With this in mind, Siemens products and solutions undergo continuous development. Siemens recommends strongly that you regularly check for product updates.

For the secure operation of Siemens products and solutions, it is necessary to take suitable preventive action (e.g. cell protection concept) and integrate each component into a holistic, state-of-the-art industrial security concept. Third-party products that may be in use should also be considered. For more information about industrial security, visit

- <http://www.siemens.com/industrialsecurity>
(<http://www.siemens.com/industrialsecurity>) To stay informed about product updates as they occur, sign up for a product-specific newsletter. For more information, visit
- <http://support.automation.siemens.com> (<http://support.automation.siemens.com>)

Disclaimer of liability

We have checked the contents of the description to ensure that it matches with the hardware and the software it describes. Nevertheless, we cannot assume responsibility for any deviations that may arise. The details outlined in this description are checked regularly and the required corrections are done in the subsequent editions. Suggestions for improvement are welcomed.

The statements in the online documentation are more binding than the statements in the manuals and PDF files. Please follow the Release Notes and Installation Notes. The information in these Release Notes and Installation Notes has priority over that in the manuals and online help about legal validity.

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Siemens AG

Digital Factory

SIMATIC Human Machine Interfaces

P.O. Box 4848

D-90026 Nuremberg, Germany.

9.2 HMI

General safety instructions

The device is designed for operation in industrial areas for operator control and monitoring of plant processes.

Open equipment and the Machinery Directive

 WARNING
The device constitutes open equipment on the back side The device constitutes open equipment on the back side. This means that the device may only be integrated in an enclosure or cabinet which provides front access for operating the device. The enclosure, the cabinet or the electrical operating rooms must provide protection against electric shock and the spread of fire. The requirements regarding the mechanical strength must also be considered. Access to the enclosure or cabinet in which the device is installed should only be possible by means of a key or tool and for trained and qualified personnel.
Electrocution risk when control cabinet is open When you open the control cabinet, there may be a dangerous voltage at certain areas or components. Touching these areas or components can cause electrocution. De-energize the control cabinet before opening it. Do not install or remove system components during operation.
The device may only be used in machines which comply with the Machinery Directive The Machinery Directive specifies precautions to be taken when commissioning and operating machinery within the European Economic Area. Failure to follow these precautions is a breach of the Machinery Directive. Such failure may also cause personal injury and damage depending on the machine operated. The machine in which the HMI device is to be operated must conform to Directive 2006/42/EC.

Observe the safety and accident prevention instructions applicable to your application in addition to the safety instructions given in the device documentation.

Safety of the plant or the system

NOTICE

Safety is the responsibility of the assembler

The safety of any plant or system incorporating the equipment is the responsibility of the assembler of the plant or system.

ESD



An electrostatically sensitive device is equipped with electronic components. Due to their design, electronic components are sensitive to overvoltage and thus to the discharge of static electricity. Note the corresponding regulations when handling ESD.

Industrial Security

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial security measures that may be implemented, please visit (<https://www.siemens.com/industrialsecurity>).

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed under (<https://www.siemens.com/industrialsecurity>).

Disclaimer for third-party software updates

This product includes third-party software. Siemens AG only provides a warranty for updates/patches of the third-party software, if these have been distributed as part of a Siemens software update service contract or officially released by Siemens AG. Otherwise, updates/patches are undertaken at your own risk. You can find more information about our Software Update Service offer on the Internet at Software Update Service (<https://support.industry.siemens.com/cs/ww/en/view/109759444>).

Notes on protecting administrator accounts

A user with administrator privileges has extensive access and manipulation options in the system. Therefore, ensure there are adequate safeguards for protecting the administrator accounts to prevent unauthorized changes. To do this, use secure passwords and a standard user account for normal operation. Other measures, such as the use of security policies, should be applied as needed.

9.2.1 Security management for HMI devices

You can find additional information on security management of HMI devices on the Internet at the following address: Panel Security Guidelines (<https://support.industry.siemens.com/cs/de/en/view/109481300>)

9.2.2 Data protection

Siemens observes the data protection guidelines, especially the requirements regarding data minimization (privacy by design). This means the following for this SIMATIC product: The product does not process / save any personal information, but only technical functional data (e.g. time stamps). If the user links this data to other data (e.g. shift plans) or if the user saves personal information on the same medium (e.g. hard disk) and therefore creates a personal reference in the process, the user has to ensure meeting the guidelines regarding data protection.

9.2.3 Notes about usage

NOTICE
The HMI device is approved for indoor use only.
The HMI device may be damaged if it is operated outdoors.
Operate the HMI device indoors only.

Note

Operate the device only in a normal atmospheric environment

The technical characteristics of the device described in the operating instructions are guaranteed if you operate the device in normal ambient air conditions with usual air composition.

Note

The device is intended for operation in an SELV/PELV circuit according to IEC/EN 61131 or IEC/EN 61010-2-201 in a dry environment, which means a dry environment on the rear of the device.

You can find additional information in the section "Operating conditions (Page 106)".

Industrial applications

The HMI device is designed for industrial applications. It conforms to the following standards:

- Requirements for interference emissions EN IEC 61000-6-4:2019
- Requirements for interference immunity EN IEC 61000-6-2:2019

Use in mixed-use zone

Under certain circumstances you can use the HMI device in a mixed-use zone. A mixed-use zone is used for housing and commercial operations that do not have a significant impact on residents.

When you use the HMI device in a mixed-use zone, you must ensure that the limits of the generic standard EN 61000-6-3 regarding emission of radio frequency interference are observed. Suitable measures for achieving these limits for use in a mixed-use zone include:

- Installation of the HMI device in grounded control cabinets
- Use of filters in electrical supply lines

Note

HMI device not intended for use in residential area

The HMI device is not intended for use in residential areas. Operation of an HMI device in residential areas can have a negative influence on radio or TV reception.

Use with additional measures

The HMI device should not be used at the following locations unless additional measures are taken:

- In locations with a high degree of ionizing radiation
- In locations with severe operating conditions, for example, due to:
 - Corrosive vapors, gases, oils or chemicals
 - Strong electrical or magnetic fields of high intensity
- In systems that require special monitoring, for example, in:
 - Elevators
 - Systems in especially hazardous areas

Notes on communication

Note

Communication errors caused by address conflict

Communication errors can occur if several devices in a network share the same bus address or IP address. Make sure that your HMI device is assigned a unique address in the network.

Note

Updating tag values following a communication error

If communication between an HMI device and controller is interrupted, all tag values displayed on the HMI device will be replaced by a hash mark ("#"). When the communication between the HMI device and controller is restored, all tag values will be updated immediately. The cycle time for updating the tag values begins again at "0".

Ethernet communication with Basic Panels with PROFINET interface

Basic Panels with PROFINET interface support the following types of communication:

- PROFINET basic function for commissioning and diagnostics
- Standard Ethernet communication

10.0 CONCLUSION

The development and integration of the Smart Mini Bottling System successfully demonstrate the application of industrial automation technologies in a real-world-inspired project. Through systematic collaboration among various teams PLC programming, SCADA/HMI design, electrical wiring, instrumentation, and system commissioning the project effectively simulated an intelligent and automated bottling line.

Despite encountering technical challenges, such as data flickering and memory conflicts in SCADA readings, the team applied practical troubleshooting techniques, including the use of MOV instructions, appropriate data types, and analog filtering methods. These solutions enhanced system stability, accuracy, and overall performance. Moreover, the SCADA and HMI systems were designed to provide a reliable, user-friendly interface for both monitoring and control, complete with alarm handling, user access management, and real-time data visualization. This project not only deepened our understanding of automation concepts but also honed our teamwork, problem-solving, and technical documentation skills preparing us for future roles in the automation and control industry.

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12.0 APPENDICES



Figure 78 : Work Progress